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PROJECT REPORT

BEng

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Numerical Analysis and Design of a Vertical Axis Wind Turbine

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Department of Engineering
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Numerical Analysis and Design of a Vertical Axis Wind Turbine

Report by
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Date
21 / 03 / 2025

DECLARATION STATEMENT

I certify that the work submitted is my own and that any material derived or quoted from the published or unpublished work of other persons has been duly acknowledged (ref. UPR AS/C/6.1, Appendix I, Section 2 – Section on cheating and plagiarism).

I confirm that any study conducted as part of my project that involved human participants has been approved in accordance with UH Ethical regulations and the protocol number(s) is/are:

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ABSTRACT

The increase in demand for sustainable energy solutions has highlighted the need for efficient small-scale wind turbines, particularly in urban and semi-urban environments. This project focuses on the numerical analysis and design of a vertical axis wind turbine (VAWT) optimised to power streetlights in semi-urban zones. The primary objective is to develop a turbine capable of operating efficiently in low and variable wind conditions, while remaining cost-effective and structurally viable for real-world implementation.

A MATLAB-based computational framework was developed to analyse wind conditions, scale turbine parameters, and estimate power output based on site-specific constraints. The effect of key design factors, including solidity, aspect ratio, Reynolds number and blade profile, were assessed to determine an optimal configuration. Two designs, a straight bladed Darrieus and a helical VAWT, were evaluated based on efficiency, manufacturing complexity, and structural performance.

Computational Fluid Dynamics (CFD) simulations were conducted using ANSYS Fluent, using 2D transient simulations with k- ω SST turbulence model to predict aerodynamic performance. A mesh independence study ensured numerical accuracy, and the results were validated against experimental data. The CFD simulations slightly overpredicted performance and saw a rightward shift in the C_p-TSR curve due to the limitations of 2D simulations. However, the trends observed in the curves aligned well with previous computational studies.

Findings indicate that the straight-bladed Darrieus turbine achieved a slightly higher annual energy production (AEP) compared to the helical design, despite the helical turbine offering smoother torque generation and better self-starting potential. Simulations to determine each turbine's cut-in speed were not conducted, therefore, the cut-in speeds were assumed constant, meaning the potential advantages of the helical design in low wind conditions remain unverified.

A cost and feasibility analysis confirmed that both turbines could meet the energy demands of a standard 30W LED streetlight, with an estimated annual energy output of around 150 kWh, and potentially turn a profit over its lifetime if installed in bulk. The straight bladed Darrieus is recommended due to higher AEP and lower manufacturing complexity and cost, whereas, the helical design is a viable alternative for installations where vibration reduction and aesthetics are primary concerns.

This study demonstrates the feasibility of using small-scale VAWTs for energy generation in urban environments. Future work could focus on high fidelity 3D CFD simulations, dynamic mesh analysis to determine actual cut-in speeds and turbine rotor speeds across different wind velocities, and experimental validation through wind tunnel or real-world testing.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Dr. MD Salim Miah, for his guidance, support, and expertise throughout this project.

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TABLE OF CONTENTS

DECLARATION STATEMENT iii
ABSTRACT iv
ACKNOWLEDGEMENTS v
TABLE OF CONTENTS vi
LIST OF FIGURES viii
LIST OF TABLES x
GLOSSARY xi
1. [Introduction] 1
1.1 [Project Aims and Objectives] 1
1.2 [Project Background] 2
1.2.1 [Horizontal-Axis Wind Turbines (HAWT)] 2
1.2.2 [Vertical-Axis Wind Turbines (VAWT)] 2
1.3 [Classification and characteristics of VAWTs] 2
1.4 [Considerations and Challenges for Small VAWTs in Urban Environments] 3
1.5 [Fundamentals of VAWT Theory and Numerical Methodologies] 3
2. VAWT Design 5
2.1 [Computational Methodology] 5
2.1.1 [Energy Requirement & Power Calculations] 6
2.1.2 [Wind Speed Predictions & Power Law Implementation] 7
2.1.3 [Weibull Distribution & Seasonal Considerations] 8
2.1.4 [Turbine Placement] 9
2.2 [Design Optimisation] 11
2.2.1 [Effects of Solidity] 11
2.2.2 [Effects of Reynolds Number] 12
2.2.3 [Blade Profile Selection & Additional Design Considerations] 13
2.3 [Initial Design] 15
2.3.1 [Evaluation of Code Outputs] 15
2.3.2 [Validation of Code Output] 15
3. CFD Simulations and Analysis 18
3.1 [Simulation Setup] 18
3.1.1 [Geometry & Domain Definition] 18
3.1.2 [Mesh Generation] 19
3.1.3 [Domain Setup & Boundary Conditions] 21
3.1.4 [Validation of Methodology] 23

3.2 [Analysis of Simulation Results] 26
3.2.1 [Blade Profile Selection] 26
3.2.2 [Effect of Solidity on Performance] 27
3.2.3 [Final Design Revisions] 29
3.2.4 [Material Selection & Cost Metric Evaluation] 31
3.2.5 [Environmental Benefits: Quantifying Emission Savings] 33
4. Project and Quality Management 35
4.1 [Project Management] 35
4.2 [Quality Management] 37
5. Conclusion 38
REFERENCES 39
BIBLIOGRAPHY 46
APPENDIX A 47
APPENDIX B 49

LIST OF FIGURES

Figure 1.1 – HAWT and VAWT representation [5] 2

Figure 1.2 – Different types of VAWTs, (A) Savonius turbine, (B) Curved-Bladed Darrieus, (C) Straight-bladed Darrieus, (D) Helical Darrieus [16] 3

Figure 1.3 – VAWT aerodynamics representation 4

Figure 2.1 - Implementation flowchart showing the codes processes for estimating turbine energy production. 6

Figure 2.2 - *Comparison of suggested mathematical approximation vs empirical data [28]* 7

Figure 2.3 - (a) Weibull distribution for January, (b) Energy production - Weibull distributed vs average wind speeds 8

Figure 2.4 - (a) Turbine placed on top of streetlight, (b) Turbine placed on separate pole, (c) Turbine placed in the middle of the streetlight 9

Figure 2.5 – (a) 180N Force applied, (b) Stress concentration on streetlight, (c) Total deformation on streetlight 10

Figure 2.6 - Required turbine size to power the streetlight at different heights from the ground. The figure is drawn proportionally to show turbine size next to a 6m streetlight. 11

Figure 2.7 – Spanwise positioning of struts (blades are blue lines, struts are black lines), (a) end struts, (b) quarter struts, (c) middle struts 13

Figure 2.8 – Basic turbine casing design 14

Figure 2.9 - Proposed helical turbine design 16

Figure 2.10 - Proposed straight bladed Darrieus design 17

Figure 3.1 - Flow domain, Mesh refinement regions (Orange lines), Interface (Purple lines), Inlet (Black lines), Outlet (Green lines), Wall (Red lines) 19

Figure 3.2 - Torque vs azimuthal angle for each mesh 20

Figure 3.3 - (a) Mesh 1 local blade refinement (b) Mesh 2 local blade refinement mesh 3, (c) Mesh outer to rotating domain 21

Figure 3.4 – (a) 5th and 6th turbine rotation torque vs azimuthal Angle, (b) Percentage change in torque vs rotation number 22

Figure 3.5 - Pressure distribution on the VAWT blades 23

Figure 3.6 - Vorticity magnitude and wake development 24

Figure 3.7 - Vorticity magnitude in the fully developed wake 24

Figure 3.8 - Cp-TSR curves against experimental data, and other computational results and comparison of both mesh types and coupled vs PISO solvers. 25

Figure 3.9 – NACA0018 vs NACA0015 C_p-TSR curves 27

Figure 3.10 – Cp-TSR curves at different solidities 27

Figure 3.11 – Straight blade vs helical turbine C_p-TSR curve 30

Figure 3.12 - Structural and mechanical components 32

Figure 4.1 – Initial Gantt Chart 36

Figure 4.2 – Revised Gantt Chart 36

Figure 0.1 - *Comparison of two reference turbines before and after solidity and Reynolds scaling* 48

LIST OF TABLES

Table 2.1– Wind shear exponent in various areas [30]: 8

Table 2.2 - Relationship between solidity and power coefficient for small VAWTs .....**Error! Bookmark not defined.**

Table 2.3 - Uncertainties in Cp through varying turbine cut-in speed..... 12

Table 2.4 – Helical turbine design parameters..... 16

Table 2.5 – Straight blade Darrieus design parameters..... 17

Table 3.1 - Effects of domain size on power coefficient 18

Table 3.2 - Validation turbine characteristics 19

Table 3.3 - Mesh independence study for mesh 2 20

Table 3.4 - Time step and rotational velocity for various TSRs..... 22

Table 3.5 - Spatiotemporal discretization schemes..... 23

Table 3.6 – AEP for different solidities..... 28

Table 3.7 – Turbine sizes for different solidities..... 29

Table 3.8 – Straight blade final design parameters..... 29

Table 3.9 – AEP comparison for straight and helical turbines..... 30

Table 3.10 – Helical turbine final design parameters 31

GLOSSARY

θ	Azimuthal angle
W	Relative velocity
U	Free stream wind velocity
V	Tangential velocity
AoA	Angle of attack
D	Drag
L	Lift
ω	Angular velocity
R	Rotor Radius
TSR, λ	Tip Speed Ratio
C_T	Torque Coefficient
A	Swept area
H	Rotor blade height
Cp	Power coefficient
ρ	Density of air
μ	Dynamic viscosity
D	Rotor diameter
P	Power
α	Hellman's wind shear exponent
h	Wind speed at height 'h'
C	Weibull scale parameter
k	Weibull shape parameter
σ_v	standard deviation of wind speed
F	Force applied by wind
P_w	Wind pressure
C_d	Drag coefficient
K	Coefficient of exposure
σ	Solidity
N_b	Number of blades
c	Chord length
Re	Reynolds number
v	Kinematic viscosity
AR	Turbine aspect ratio
Γ	Gamma function

C_f	Skin friction coefficient
τ_w	Wall shear stress
u_τ	Wall shear velocity
y	First layer height
y+	Dimensionless wall distance
t	Time step size
ROI	Return on investment
AEP	Annual energy production

1. [Introduction]

As global energy demand continues to rise, the transition from fossil fuels to renewable energy sources has become a critical priority. The combustion of fossil fuels for electricity is one of the largest contributors to greenhouse gas emissions, accounting for approximately 40% of global emissions [1]. In response, international agreements such as the Paris agreement aim to limit global temperature rise by reducing carbon emissions and increasing reliance on sustainable energy sources.

Urban environments in particular present a significant challenge for renewable energy adoption. Cities are responsible for nearly 75% of global energy consumption and 70% of greenhouse gas emissions. Outdoor urban lighting alone can account for as much as 20 – 40% of municipality's electricity budget expenditure [2], making it a prime candidate for energy efficiency and economic improvements. Many cities have adopted LED streetlights to reduce electricity consumption, there remains an opportunity to further enhance sustainability by transitioning streetlight power sources from grid electricity to renewable generation.

Wind energy is one of the most promising renewable energy sources due to its high availability and scalability. The UK holds around 40% of Europe's total wind energy [3]. However, traditional wind turbines such as large horizonal axis wind turbines (HAWTs) are not well suited to urban environments sue to space constraints, noise concerns and performance in the turbulent wind conditions caused by buildings and other structures. Vertical axis wind turbines (VAWTs) offer a practical solution as they can operate efficiently in varying wind conditions [4].

This project focuses on the design, analysis and optimisation of a small-scale VAWT intended to power streetlights in semi-urban areas. Through use of computational fluid dynamics (CFD) simulations, this project aims to develop a turbine capable of generating sufficient power under low wind and variable conditions.

1.1 [Project Aims and Objectives]

The main aim of this project is design, simulate, and evaluate a small vertical axis wind turbine which can provide a reliable power source for streetlights in semi-urban areas. The goal is to create a design that not only works theoretically but can also be practically implemented in semi-urban areas.

Objectives:

- Develop a Vertical axis wind turbine which works effectively in environments where wind speeds can be low and unpredictable. Explore different features that can help the turbine capture as much energy as possible in semi urban zones, to reliably power streetlights during low wind speed periods.
- Optimize turbine design for performance and cost effectiveness; the design will be tailored to perform in variable and low wind conditions. Through detailed CFD analysis of key turbine parameters such as solidity, aspect ratio, helical angle, general turbine geometry and dimensions to determine a suitable turbine configuration for the chosen site.
- Assess the overall feasibility of the turbine design; perform a cost analysis which covers material, manufacturing, installation and maintenance expenses. Practical strategies for installing and using these turbines in urban environments will also be recommended. Quantify environmental benefits from the proposed design.

Research approach and measuring effectiveness:

This project will combine theoretical analysis, CFD simulations, cost and feasibility evaluations. The effectiveness of the turbine design will be measured by its performance in terms of efficiency (C_p-TSR curves), estimated annual energy production, and overall cost effectiveness in comparison to current

methods of powering streetlights. By following these steps, the project aims to deliver a practical and well-optimised turbine design for powering streetlights in semi-urban zones.

1.2 [Project Background]

This section summarises key wind turbine types, their operating principles, and their respective advantages and limitations.

1.2.1 [Horizontal-Axis Wind Turbines (HAWT)]

HAWTs have their rotors aligned with the wind and have been extensively developed. They have high aerodynamic efficiency, often reaching 75 – 80% of the Betz limit, making them popular for large wind farms. However, HAWTs need unobstructed wind flow, complex yawing to face the direction of the wind and self-starting mechanisms, and regular high altitude maintenance. These factors make them less suitable for urban or semi-urban settings and closely spaced installations, such as those required for streetlights. [4]

1.2.2 [Vertical-Axis Wind Turbines (VAWT)]

VAWTs have rotors mounted perpendicular to the wind, making them well-suited for environments with variable wind directions and limited space. Key components such as the gearbox and generator can also be installed at ground level, VAWTs are also easy to maintain and less noisy. Although they generally have lower aerodynamic efficiency due to pulsating torque from changing angles of attack, their simplicity and versatility make them ideal for small scale urban applications. [4]

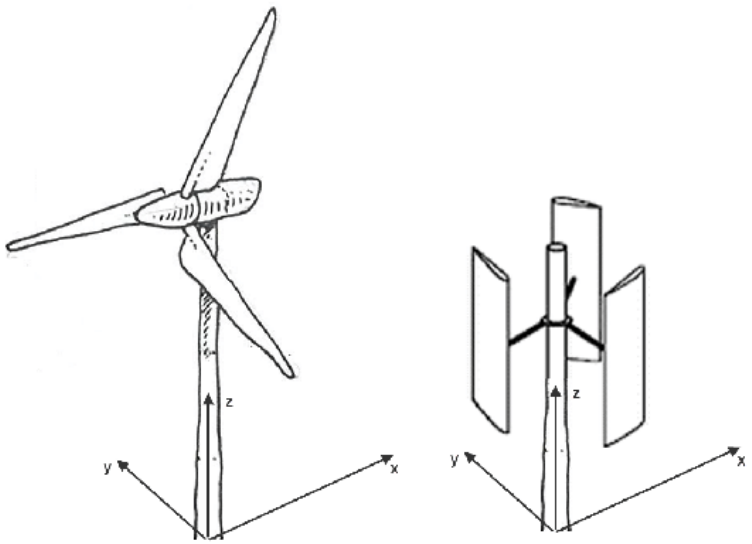


Figure 1.1 – HAWT and VAWT representation [5]

1.3 [Classification and characteristics of VAWTs]

VAWTs come in several designs, each with unique benefits and drawbacks.

Savonius Turbine: Developed in the 1920s, Savonius turbines used curved ‘bucket’ blades to generate torque primarily through drag. Their excellent self-starting ability and performance in turbulent, multi-directional winds come at the cost of low efficiency, with typical power coefficients of around 0.15 (capturing only 15% of the wind energy) [6], [7], [8]. This low efficiency limits their use to lower power applications and wind-speed measurements [4].

Darrieus Turbine: Darrieus turbines, introduced in the 1930s, use lift generated by airfoil shaped blades to produce torque and typically produce power coefficients between 0.25 and 0.35 [7]. A

notable drawback is their poor self-starting performance, increasing turbine solidity can increase self-starting ability, however, could reduce overall efficiency [9], [10].

Helical and Modified Darrieus Designs: Helical designs twist the blades along their length to smooth out pulsating torque, reduce vibration, and improve self-starting ability, though their maximum power coefficient is typically slightly lower than straight bladed designs. [11], [12]. Other modifications, such as J-bladed and curved configurations, aim to enhance self-starting and provide good efficiency but are generally more complex and costly to manufacture [13], [14].

Hybrid Designs: Hybrid turbines combine elements from both Savonius and Darrieus models, using the Savonius component for strong starting torque and the Darrieus for higher efficiency. The challenge lies in balancing these elements, if misaligned, the drag forces may induce negative torque [15]. Hybrid designs often achieve higher overall performance characteristics in many numerical studies by leveraging both turbine designs, however, at a small cost of Cp [4].

Common types of VAWTs are shown in *Figure 1.2*:

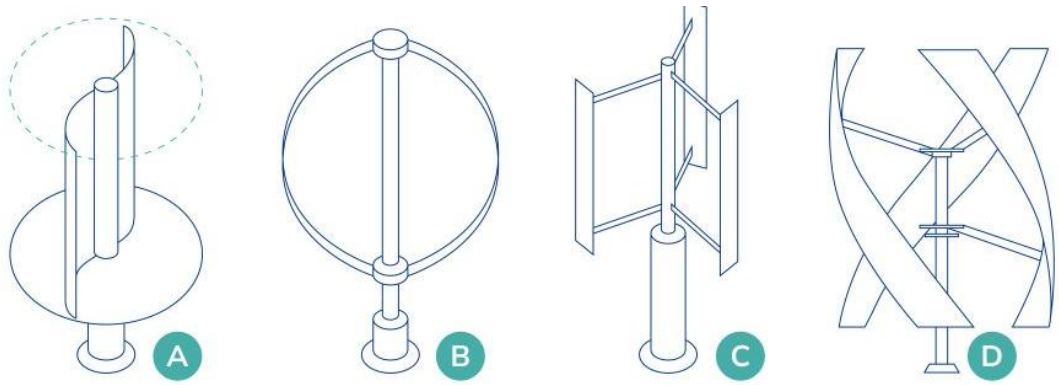


Figure 1.2 – Different types of VAWTs, (A) Savonius turbine, (B) Curved-Bladed Darrieus, (C) Straight-bladed Darrieus, (D) Helical Darrieus [16]

1.4 [Considerations and Challenges for Small VAWTs in Urban Environments]

Urban wind conditions are often turbulent and inconsistent due to building induced obstructions. This necessitates designing small VAWTs with appropriate solidity, aspect ratios, and blade geometries to perform reliably in low wind speeds. Helical and hybrid designs offer improved self-starting and smoother operation, which can make them more beneficial for urban settings [4].

Structural integrity, noise levels, and maintenance are crucial. Savonius turbines are simple and low cost, require larger sizes to generate sufficient power, whereas, Darrieus turbines, despite being more compact can suffer from fatigue due to cyclic loading [17]. Economically, the turbine must balance efficiency with installation and maintenance costs to be viable for applications like streetlighting [18].

Wind energy systems, including small VAWTs, generally have lower carbon footprints than fossil fuel alternatives. Life cycle assessments typically show over a 20-25 year life span offer favourable energy payback and reduced greenhouse gas emissions [19]. The use of lightweight, recyclable materials further enhance resource efficiency and supports broader environmental social and environmental benefits, such as improved air quality and job creation

1.5 [Fundamentals of VAWT Theory and Numerical Methodologies]

Figure 1.3 illustrates the aerodynamics of a three-bladed VAWT, where the blades continuously encounter varying wind conditions as they rotate around a vertical axis. The theory is based on [20].

During the upwind portion (θ = 0° - 180°, where θ is the azimuthal angle), the blade moves against the incoming wind, resulting in a high relative velocity (W) as the rotational velocity (V = ωR) adds to the free-stream wind (U). This effect is most pronounced near θ = 90°, where favourable angles of attack generate significant lift forces. In contrast, the downwind cycle (θ = 180° - 360°), blades enter the wake of the upwind section and experience reduced relative velocity and lower angles of attack, therefore generating less lift.

VAWTs generate pulsating torque due to the cyclical variations in aerodynamic forces, a challenge which isn't normally seen with the steadier output of HAWTs. Dynamic stall also may occur when the angle of attack (α, AoA) exceeds the airfoils stall angle which can further compromise performance. Wake interactions between blades also contribute to turbulence and reduced kinetic energy which can affect overall efficiency.

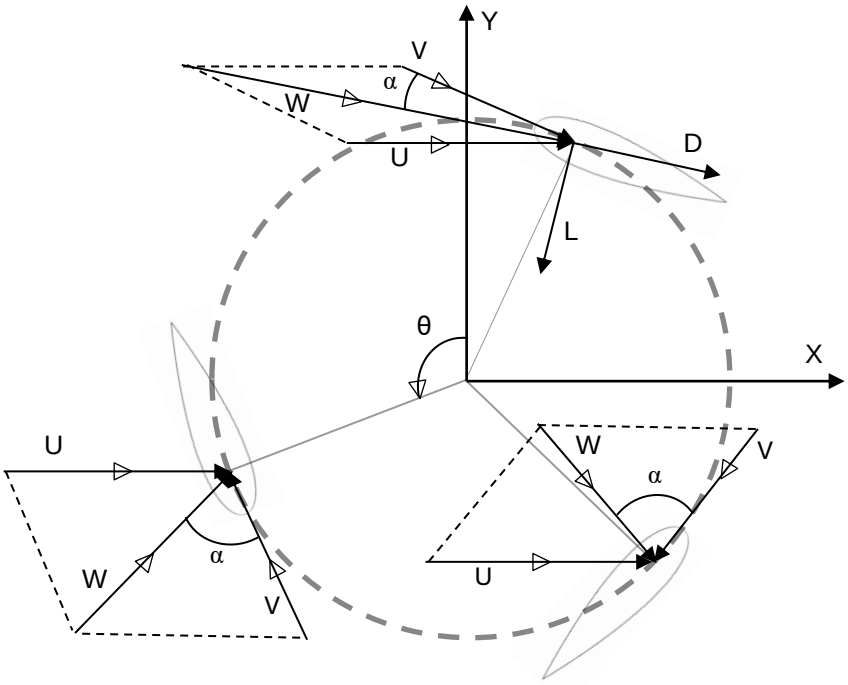


Figure 1.3 – VAWT aerodynamics representation

L refers to the lift force which acts perpendicular to the relative velocity and D is the drag force which acts parallel to the relative velocity.

Tip speed ratio (λ, TSR) refers to the speed of the blades relative to the free-stream wind speed, and is a crucial parameter in VAWT design. TSR can be expressed as:

$$\lambda = \frac{V}{U} = \frac{\omega R}{U}$$

Equation 1.1

2. VAWT Design

2.1 [Computational Methodology]

A MATLAB code was developed to suggest initial turbine geometry for powering streetlights based on specific input parameters. Methods such as BEM or DMSTM can offer more detailed power predictions they are computationally expensive and don't capture full 3D effects (e.g. for helical blades).

The code calculates wind speeds using an average wind speed and wind index, estimates the optimal TSR and power curve based on solidity and other inputs, then approximates turbine output power. Different VAWT configurations are compared by applying Reynolds and solidity scaling to determine the minimum required dimensions (chord length, rotor radius, and blade height) to meet a specific power output. Although the code serves as a good baseline, further optimisation and validation of the codes initial turbine suggestion is recommended through CFD.

The workflow is as follows (see *Figure 2.1* for the flowchart)

1. Initialization: Define reference turbine properties (C_p, TSR, Re, blade number, solidity, cut-in speed) along with the site parameters (wind shear exponent, average wind speed, turbine height, monthly wind index).
2. Wind speed Calculation: Compute monthly wind speeds based on Weibull distribution and power law.
3. Energy demand calculations: The energy demand for streetlights is calculated based on daylight hours each month and streetlight wattage in London, UK.
4. Scaling: Scale the turbine's C_p using empirical data as a function of solidity and Reynolds number.
5. Optimization: Loops through rotor radius' and aspect ratios to calculate turbine dimensions and verify production meets the energy requirement.
6. Plotting and reporting: Generates plots to compare turbine outputs, validates methodology and summarizes results.

Typical efficiency losses such as mechanical losses (0-0.3%), electrical losses (1-1.5%), downtime (failure or maintenance, 2-3%) and efficiency losses of around 3-10% for transferring energy to the grid, however the energy in this case is fed directly to the streetlight negating this loss [21] were modelled within the MATLAB code.

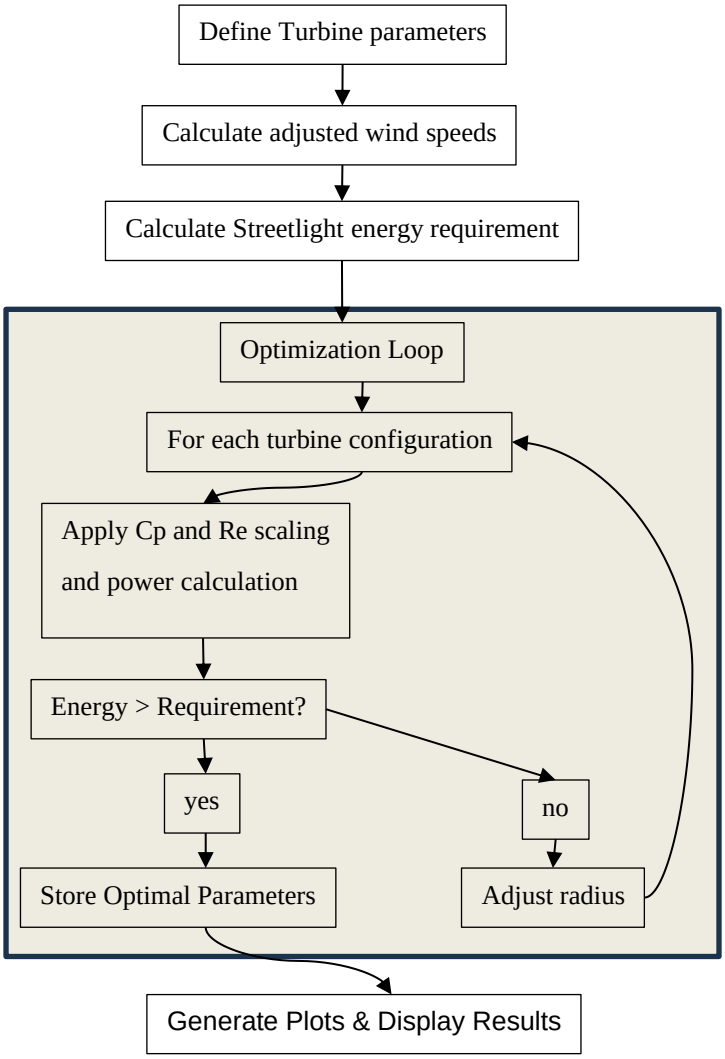


Figure 2.1 - Implementation flowchart showing the codes processes for estimating turbine energy production.

2.1.1 [Energy Requirement & Power Calculations]

The streetlight's energy requirement is calculated using monthly daylight hours in London, UK [22]. This monthly approach is more precise than using an annual metric such as Elexon's burning hours and is crucial for assessing the practicality of the VAWT design, the streetlight energy requirement is calculated by multiplying the streetlights operating hours by the lamps wattage. Based on data from Camden, UK [23] streetlight wattages range from 3W-150W, an average value of 30W (typical for LED streetlights) is used. LED lamps are targeted as they have better energy efficiency and a lower carbon footprint than other lamps, which aligns with the sustainability goals of this project whilst ensuring the size of the turbine stays realistic.

The power output of a VAWT is expressed by:

$$P = \frac{1}{2}\,\rho\, V_0^3\, D\, H\, C_p$$

Equation 2.1

Where ρ is the density of air, V_0 is the wind velocity, D is the turbine diameter, H is the turbine height, C_p is the turbines power coefficient and P is the output power. It should be noted that C_p is limited by the Betz limit (59.3%)[24], which serves as a benchmark for realistic turbine performance.

To accurately model the turbine operating away from its maximum C_p the C_p-TSR curve was approximated using empirical relationships and mathematical formulation. A detailed breakdown of the methodology is provided in Appendix C. Figure 2.2 shows the approximation against the empirical data. The approximation aligns well for turbines with solidity above 0.08, however, further refinement may be needed for lower solidity designs. Overall, this gives confidence in the MATLAB codes ability to predict C_p at different wind speeds.

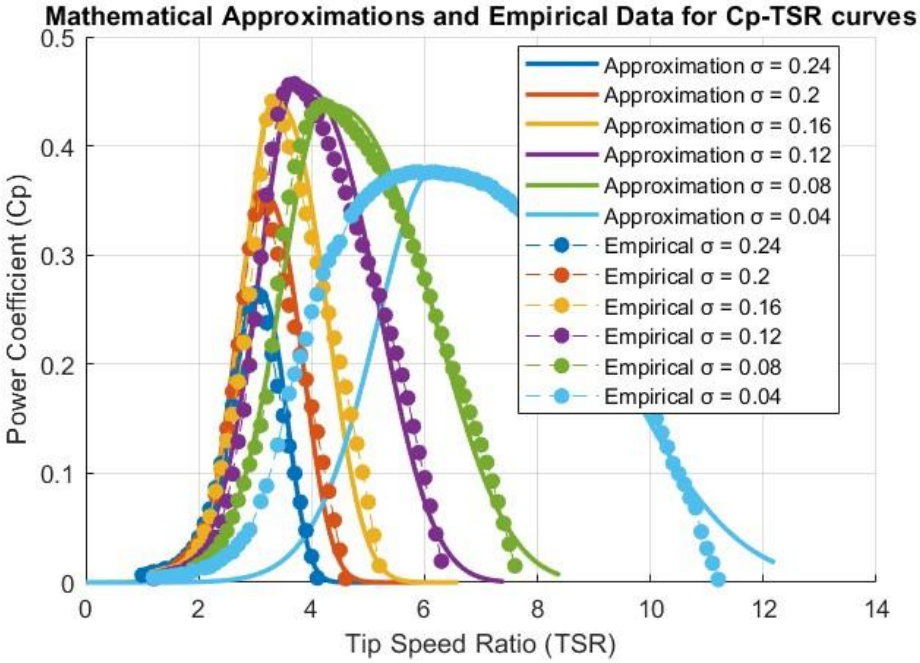


Figure 2.2 - *Comparison of suggested mathematical approximation vs empirical data [25]*

2.1.2 [Wind Speed Predictions & Power Law Implementation]

Site wind speeds are estimated through use of Global Wind Atlas [26]. A 9 km^2 area is made around Royston, UK, which is a typical semi-urban area, and the average wind speed is taken from the bottom 20% of windiest zones to ensure the turbine is effective across the majority of semi-urban areas. This gives an average wind speed of 4.7 m/s at a reference height of 10m.

Wind speeds at different heights can be calculated using the power law:

$$V(h) = V_{ref}\left(\frac{h}{h_{ref}}\right)^{\alpha}$$

Equation 2.2

Where V_{ref} is measured wind speed at reference height h_{ref}, α is the Hellman's wind shear exponent and V(h) is the wind speed at height h.

In urban and semi-urban environments, increased surface roughness due to buildings can result in higher shear exponents. Based on typical conditions for semi-urban areas (see *Table 2.1* [27]), an α value of 0.35 is chosen to reflect these effects.

Area Type	α
Smooth hard ground, calm water	0.1
Tall grass on level ground	0.15
High crops, hedges and shrubs	0.2

Wooded countryside, many trees	0.25
Small town with trees and shrubs	0.3
Large city with tall buildings	0.4

Table 2.1– Wind shear exponent in various areas [27]:

2.1.3 [Weibull Distribution & Seasonal Considerations]

To account for the variability of wind speeds, which is critical as turbine output scales cubically with wind speed, the Weibull distribution is used. This model offers several benefits:

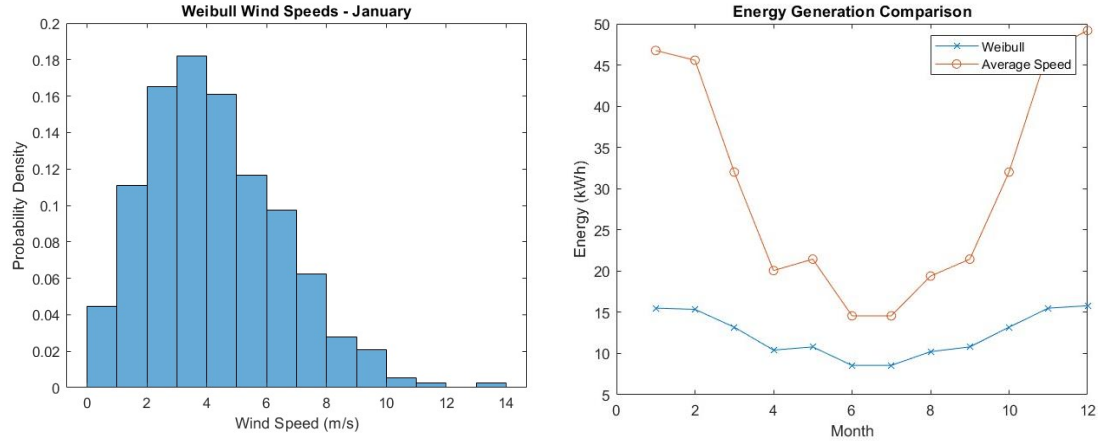
- Accurate Variability Capture: It reflects both frequent low speeds and infrequent high speeds, therefore avoiding inaccuracy which can occur when relying on average wind speeds.
- Adaptability to Urban Conditions: With the adjustable shape (k) and scale (C) parameters, the distribution can model different wind speed regimes. Urban areas typically have higher turbulence and obstructions and tend to show lower k values. For example, Fruh's analysis [28] of UK stations show k values ranging from 1.43 to 2.23. For a built up area like Heathrow k ≈ 2.03 at 25m, after adjusting for turbine height at around 7m a k value of 1.9 is chosen for forthcoming calculations.

Limitations:

- The model does not incorporate turbulence into the wind speed predictions, however, this will be explored in later ANSYS simulations when higher accuracy is needed.
- General assumptions of scale, shape and wind shear exponent parameters can lead to inaccurate results in that specific area however, for the wind turbine to power streetlights in a variety of semi-urban zones a generalized approach can be beneficial.

Appendix B shows the mathematical formulation of the Weibull distribution used in the code.

Seasonal variations are accounted for using monthly wind indices from Global Wind Atlas. These variations are crucial for realistic turbine performance calculations. For example, wind speeds are typically lowest in June/July which coincides with reduced streetlight energy demand due to longer daylight hours which therefore mitigates some concerns about production over low wind periods.



(a) (b)

Figure 2.3 - (a) Weibull distribution for January, (b) Energy production - Weibull distributed vs average wind speeds

*Figure **2.3*** (b)'s results show annual energy production with Weibull wind speeds being 60% lower overall than using average speeds. This reduction reflects the realistic losses from below cut-in

conditions, non-optimal C_p values (modelled by the C_p-TSR curve), and mechanical inefficiencies all of which are particularly important in urban conditions.

2.1.4 [Turbine Placement]

UK streetlights vary in height (5-12m), a 6m streetlight is chosen here due to the lower wattage of the chosen lamp [29]. *Figure 2.4* shows three possible configurations for where the turbine could be placed relative to the streetlight.



(a) (b) (c)

Figure 2.4 - (a) Turbine placed on top of streetlight, (b) Turbine placed on separate pole, (c) Turbine placed in the middle of the streetlight

Configuration (a) captures the most wind energy as it is mounted higher therefore, being subjected to higher wind speeds, it also provides less obstruction, however, may sustain higher bending moments. Configuration (b) reduces bending moments, however, has lower energy capture and may experience wake effects from the streetlight. Configuration (c) also has low energy capture and may obstruct the light source. Economically, configuration (a) is advantageous as it avoids the extra cost of installing a separate pole or modifications for mid pole mounting.

To address the main concern of configuration (a), its structural feasibility was assessed using basic finite element analysis (FEA). The streetlight dimensions and materials were modelled acuratey, and the turbine and casing were modeled as a high stiffness cylinder as streetlight deflection and failure are the main focus of the analysis.

The force applied by the wind can calculated using methodology presented in [30]:

$$F = A * P_w * C_d$$

Equation 2.3

Where F is the force applied by the wind, A is the area of the structure, P_w is wind pressure and C_d is the drag coefficient of the structure. For a more accurate FEA result, CFD simulations could be carried out to determine the drag coefficient of the configuration. However, due project time constraints a standard drag coefficient for a cylinder will be used instead (C_d = 1.2).

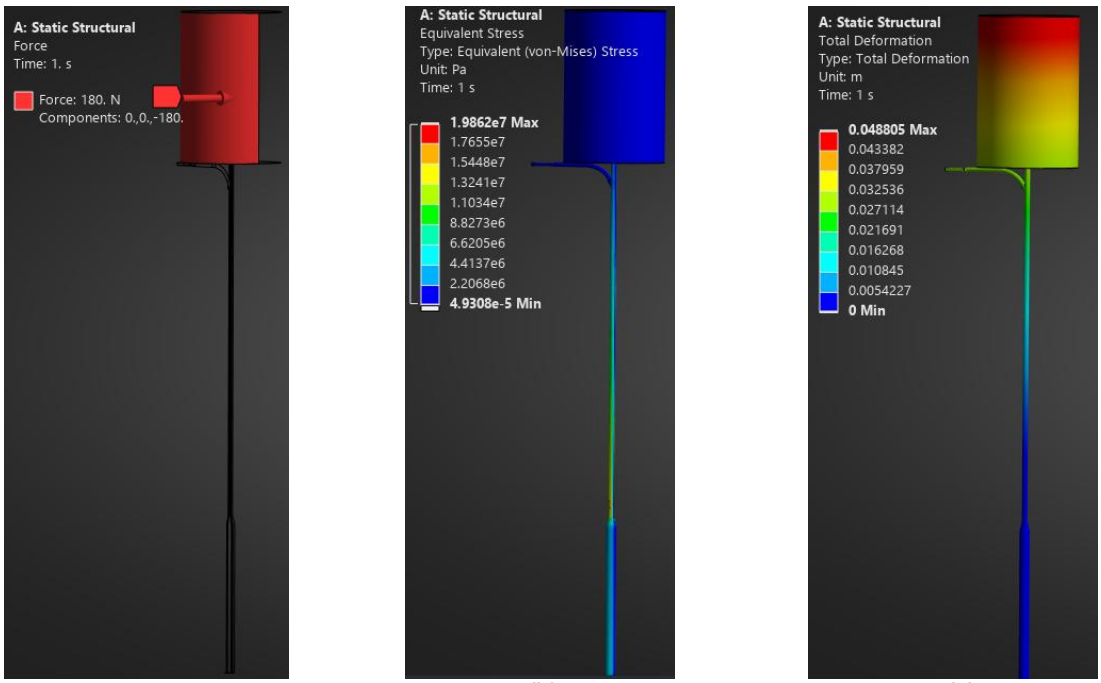
The wind pressure can be estimated using *Equation 2.4*:

$$P = K * U^2 * 1.15$$

Equation 2.4

Where 'K' is the coefficient of exposure and U is the wind speed. The 1.15 is multiplied as recommended by best practices for tall and slender structures. For an urban area with structures under 70 feet, K is estimated as 0.00094.

At a wind speed of 14 m/s, a force of 180N is uniformly applied across the turbines swept area (*Figure 2.5* (a)), giving a maximum stress of approximately 20MPa (*Figure 2.5* (b)). Given that steel streetlights typically have a yield strength between 250 and 600MPa [31], [32] and applying a safety factor of 4 [33] (as civilians could be near at any time) an allowable stress of 42 MPa is obtained, therefore configuration (a) can be considered structurally safe.



(a) (b) (c)

Figure 2.5 – (a) 180N Force applied, (b) Stress concentration on streetlight, (c) Total deformation on streetlight

Configuration (a) may require more frequent inspections whilst also being harder to access due to increased turbine height, the increased cost of the higher altitude vehicle has been factored into the cost analysis in later sections. Configuration (a)'s better wind capture and lower initial costs should outweigh these extra maintenance concerns.

Figure 2.6 shows that achieving the required energy output (125kWh per year for a 30W streetlight) at lower heights would need a larger turbine swept area. (The method to estimate the swept area is based on basic theoretical calculations and assumptions shown in section 2.3.2)

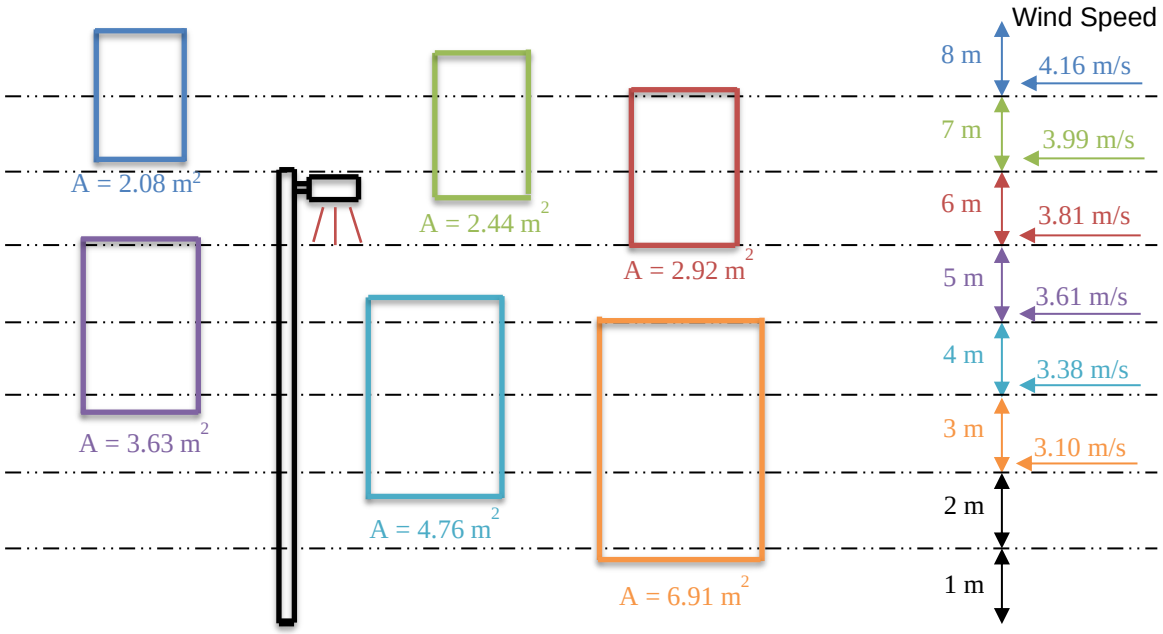


Figure 2.6 - Required turbine size to power the streetlight at different heights from the ground. The figure is drawn proportionally to show turbine size next to a 6m streetlight.

Based on these trade-offs, configuration (a) is chosen as the final turbine placement, however, more detailed structural and cost analysis can be done in future work.

2.2 [Design Optimisation]

This chapter focuses on the aerodynamic and structural factors influencing turbine performance.

2.2.1 [Effects of Solidity]

Turbine efficiency is highly sensitive to solidity, as it influences the C_p-TSR curve and overall performance across varying wind speeds.

Solidity (σ) can be defined by *Equation 2.5* [34]:

$$\sigma = \frac{N_b c}{R}$$

Equation 2.5

Where N_b is the number of blades, R is the rotor radius and c is the blades chord length.

However, some literature [35] defines it as:

$$\sigma = \frac{N_b c}{2R}$$

Equation 2.6

This difference is critical as it changes the definitions of high and low solidity and therefore, turbine design parameters such as chord length. The report relies on the latter formula and so all empirical solidity values are converted accordingly.

Higher solidity increases the interaction between the blades and airflow due to larger blade surface area therefore, enhancing C_p and starting torque at low wind speeds. At higher wind speeds, increased drag can reduce C_p whilst also demanding more material and weight. Typically, high solidity (σ = 0.3+) turbines operate at low TSRs whilst low solidity (σ = 0.05-0.15) at higher TSRs.

With the design wind speed being 3-4 m/s at turbine height, the proposed turbine is designed with higher solidity, therefore, the turbine will likely operate at an optimal TSR of 1-3. The higher solidity enhances self-starting ability in low-wind, urban or semi-urban conditions. [36], [37]

To capture the effects of solidity a scaling process was created in the MATLAB code based on empirical data, the detailed breakdown of this is shown in Appendix A.

Changing solidity also affects the Reynolds number (a factor addressed in the following section) and can also influences a turbines cut-in speed. Higher solidity turbines typically show lower cut-in speeds by increasing starting torque, whereas lower solidity can show higher cut-in speeds. Due to limited data on the relationship between cut-in speed and solidity, the cut-in speeds are estimated based on literature. Due to the uncertainty this brings a sensitivity analysis was performed to measure its impact. As shown in *Table 2.2*, increasing the cut-in speed from 2.5 m/s to 4 m/s (realistic cut-in speeds [38]) results in a total energy drop of up to 16.24%. This highlights the need for designing turbines with lower cut-in speeds in urban and low-wind environments.

Cut-in Speed (m/s)	Energy (kWh)	% Change in Output power	Cumulative % Change in Output Power
2.5	149.61	0.00%	0.00%
3	147.91	-1.13%	-1.13%
3.5	139.27	-5.76%	-6.89%
4	125.24	-9.35%	-16.24%

Table 2.2 - Uncertainties in Cp through varying turbine cut-in speed

Bianchini, A et al. [39] suggests that for wind speeds around 4 m/s an optimal solidity of 0.47-0.51 (using NACA0015 or NACA0018 profiles) is recommended for turbines with similar swept areas and aspect ratios to the proposed design. The initial turbine geometry will be designed with 0.4 solidity, however, different solidities will be tested in the CFD stage.

2.2.2 [Effects of Reynolds Number]

Reynolds number can significantly affects a turbines' performance as it describes the flow regime around the blade. Small VAWTs in urban environments typically have Re values of below 100,000 where flow separation is frequent.

The local, chord based Reynolds number can be defined as:

$$Re_c = \frac{c\,w}{v}$$

Equation 2.7

Where c is the chord length, w is the air speed relative to the airfoil and v is the kinematic viscosity of air. Following S. Brusca's [40] methodology w is approximated by ωR and is done for an averaged Reynolds number independent of AoA. Given that VAWT rotational velocity increases with wind speed until aerodynamic regulation maintains constant ω, this approximation can be justified [41].

ωR can be described by:

$$\omega \mathrm{R} = \mathrm{TSR}_{\mathrm{opt}} V_0$$

Equation 2.8

Combining *Equation 2.7* and *Equation 2.8* gives:

$$Re_c = \frac{c\, V_0 TSR_{opt}}{\nu}$$

Equation 2.9

Chord based Reynolds number is used as the turbines power coefficient is heavily affected by blade geometry. Research by Mark Aaron Miller et al. [42] shows that C_p increases significantly with Re_c in the range of 10^4 – 10^5 before gradually leveling off.

Since solidity directly influences the chord length, the Re is scaled based on the updated solidity scaled chord length. The newly calculated Re using the adjusted chord length, TSR and wind speed, is then compared to a reference value. A scaling factor for C_p is then found by interpolating from the experimental C_p-TSR curve [42], therefore, ensuring that the impact of Reynolds number variations on turbine performance is captured.

The MATLAB-based scaling methodology was validated against experimental data and showed good alignment, making it suitable for analysis. The full results and methodology are outlined in Appendix A

2.2.3 [Blade Profile Selection & Additional Design Considerations]

Airfoil selection is an important factor in the design of a VAWT as it influences the turbines efficiency, self-starting and structural properties of the turbine. Higher solidity VAWTs operate at lower TSRs where high AoA causes stall and flow separation, therefore, airfoils with high lift coefficients, stall angles and good post stall performance should be chosen. Laminar airfoils may achieve lower cut-in speeds however, have less efficiency during operation, whereas, airfoils designed for turbulent flows often require a starting mechanism [43]. Maleki et al. [44] demonstrated that although cambered airfoils offer approximately 40% better starting ability symmetric airfoils can have up to a 506% increase in power coefficient. Wang, Y et al [45] recommends NACA0018 for TSRs below 2.33 and NACA0015 for higher TSRs. The proposed turbine design will likely have an optimal TSR of 1-3 therefore, both airfoils will be analysed in CFD simulations to determine the best blade profile for the proposed design.

Strut Design and Aspect Ratio

Strut configuration plays a significant role in both structural and aerodynamic efficiency. Poorly designed struts could lead to efficiency losses of up to 50% this effect is also more prominent at higher TSRs. For turbines operating at lower TSRs the commonly used airfoil shaped struts may not be necessary. However, strut placement is a crucial consideration to mitigate losses, *Figure 2.7* shows the different spanwise connections. The spanwise connections are commonly made at the middle, quarter or end span. While middle-span connections minimize drag, end struts tend to show the least reduction in efficiency as they reduce tip losses and don't affect the middle high performance section of the blade, however, can experience up to five times higher stresses compared to quarter span struts. Quarter span placement at around 20.7% of blade length from each tip [46], offers a balanced aerodynamic and structural compromise. Chordwise, struts are typically attached at 0.25c to align with the airfoils aerodynamic centre, as opposed to mid chord placement which can cause unwanted blade forces, especially with high solidity VAWTs [47]. A standard strut width of 0.5c is a reasonable choice as it is commonly used in VAWT design and should withstand the varying wind speeds in urban environments.

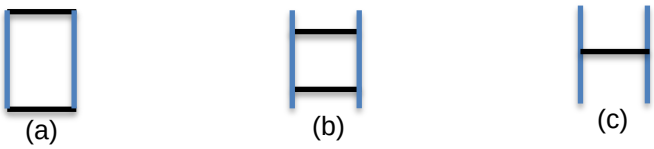


Figure 2.7 – Spanwise positioning of struts (blades are blue lines, struts are black lines), (a) end struts, (b) quarter struts, (c) middle struts

Aspect ratio can be defined as:

$$AR = H/R$$

Equation 2.10

Where H is blade height and R is rotor radius.

Aspect ratio is a crucial design parameter. Smaller AR increases chord length (*Equation 2.5*), and therefore the Reynolds number (*Equation 2.9*), which would intuitively increase power coefficient based on [42], however, it also increases blade tip losses. Research by Zanforlin et al. [48] and Paraschivoiu [49] recommends an optimal aspect ratio between 2.6 and 3, which balances aerodynamic performance and structural efficiency, the larger aspect ratio also mitigates some concerns of the turbines horizontal spatial footprint.

Additional Design Considerations

Wind unpredictability means the inclusion of an energy storage system to ensure consistent power delivery to streetlights. Storing a few days' worth of energy can buffer low-wind periods whilst being inexpensive and spatially feasible to implement. Sizing the turbine to meet annual energy demand alone could lead to undersupply in low wind months. To account for this the turbine is designed to cover the worst case monthly deficit in energy, this leads to around 20% extra annual energy production. However, storing all this excess energy is not feasible as storage systems would be too large and expensive to integrate (which is the same problem as designing for the annual streetlight requirement and storing energy from windier months to power months with lower wind). Alternatively, integrating an inverter to feed excess energy back to the grid could allow for a smaller turbine design however, at increased complexity and cost.

A protective casing has also been designed to guard the turbine against debris, extreme weather, and unauthorised access. More advanced casing designs with guide vanes can theoretically boost C_p by up to 35% [44], however, due to cost constraints a simpler casing such as that shown in *Figure 2.8*, that minimally impedes airflow while providing necessary protection is designed.

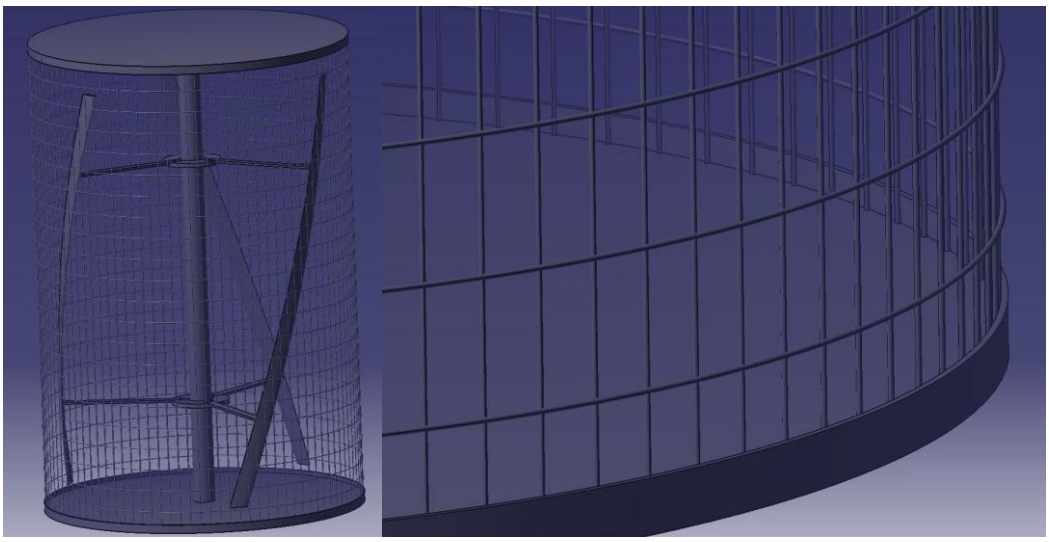


Figure 2.8 – Basic turbine casing design

To prevent damage to turbine components, especially the blades and generator, wind turbines must be designed with a maximum operational wind speed. Wind speeds above a certain threshold can result in mechanical stress, overheating of the motor, and potential failure of parts due to increased aerodynamic forces. A simple mechanical brake system is implemented which slows the rotor when sensors detect the rotational velocity is too high.

The turbine generates power through a generator. The turbine shaft rotates on ball bearings inside the case, which turns the motor to generate power. The motor should be capable of generating power of the maximum possible wind energy the turbine will extract from the wind. This can be calculated using the Betz limit (59.3%), *Equation **2.1***, a maximum turbine operational wind speed of 10 m/s, and an estimated swept area of around 3 m^2, this comes out to roughly 1 kW (0.5 * 3 m^2 * 1.225 kg/m^3 * (10 m/s)^3 * 0.593 = 1.089 kW). Therefore, a 1kW motor should be used for this design.

2.3 [Initial Design]

2.3.1 [Evaluation of Code Outputs]

The code identified the helical turbine as the design with the smallest swept area capable of achieving the required power output, which is important in urban or space constrained environments. The straight blade Darrieus has a slightly larger swept area, however, benefits from a simpler geometry that reduces manufacturing complexity and cost.

These findings align with suggestions in literature. For example, Shubham S et al. [50] recommends a 2-3 bladed straight bladed Darrieus with a large aspect ratio and a mid-range solidity for low wind speed areas (Nottingham 2.57 m/s wind speed), whereas, Fleisinger, M et al. [10] suggests a helical design may be better due to the straight blade Darrieus's reduced performance in turbulent conditions and poor self-starting ability.

Trade-Offs Between the Helical and Straight-Bladed Designs

1. Spatial Footprint and Aesthetics:
    - Helical Turbine: Offers a compact swept area and aesthetic design that is less obstructive in confined areas.
    - Straight-Bladed Darrieus: Requires a larger swept area but its simple blade geometry may be easier to integrate with existing infrastructure.
2. Aerodynamic and structural Performance:
    - Helical Turbine: The continuous twist of the blades provides smoother aerodynamic loading over a full rotation which could potentially reduce vibrations and stresses [51], potentially leading to lower maintenance costs over the turbine's life.
    - Straight-Bladed Darrieus: May experience less smooth loading which can lead to higher stress levels and increased noise during operation.
3. Manufacturing Complexity and Cost:
    - Helical Turbine: The complex geometry can increase production costs and time due to the need for specific manufacturing methods.
    - Straight-Bladed Darrieus: The simpler design allows for more regular manufacturing processes, which can reduce manufacturing costs and increase its scalability [52].
4. Installation and Maintenance:
    - Helical Turbine: The complexity of the blade geometry might require more detailed and specific maintenance procedures.
    - Straight-Bladed Darrieus: The simple geometry could lead to easier installation and maintenance routines, which would be beneficial in built up areas.

2.3.2 [Initial Designs]

The generated code outputs were validated mathematically as shown in Appendix D and show good alignment with theoretical expectations. In summary, the MATLAB code can be used as a framework to simplify the initial design process by making it easy to balance aerodynamic performance and operational conditions. Further development for higher wind speed designs can also be carried out if needed. The helical turbine is recommended where minimising spatial footprint and noise and structural feasibility is concerned, whereas, the straight-bladed Darrieus is recommended where minimising cost and maintenance simplicity is critical.

Both designs were modelled in CATIA (*Figure 2.9* and *Figure 2.10*) to provide a visual representation and allow for further analysis in ANSYS Fluent. Turbine dimensions are also shown in *Table 2.3* and *Table 2.4*.

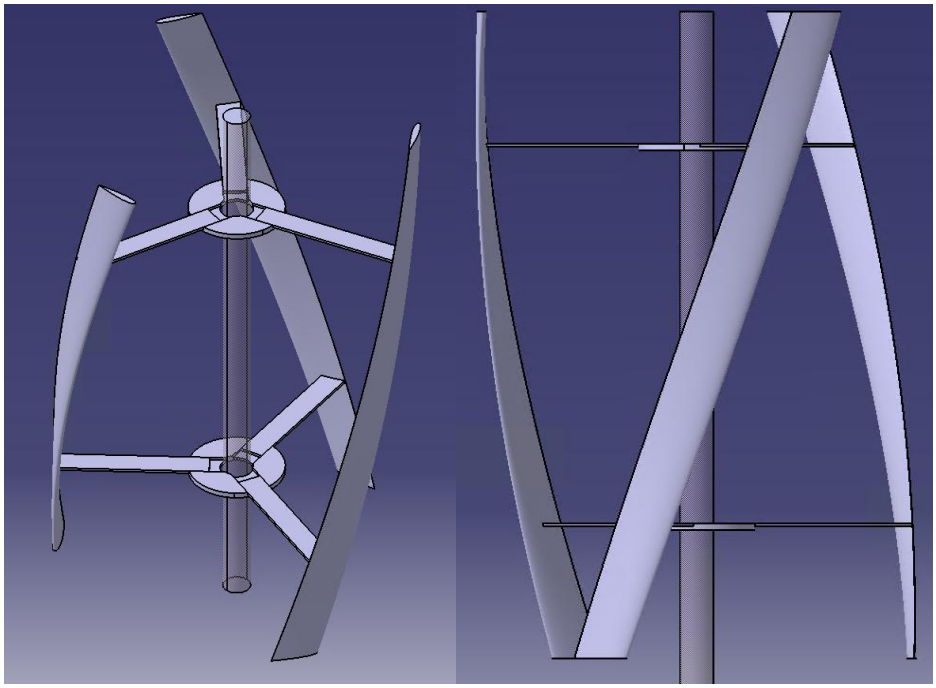


Figure 2.9 - Proposed helical turbine design

Blade Height	1.98 m
Rotor Diameter	1.32 m
Chord Length	0.176 m
Helical Angle	60 Degrees
Solidity	0.4

Table 2.3 – Helical turbine design parameters

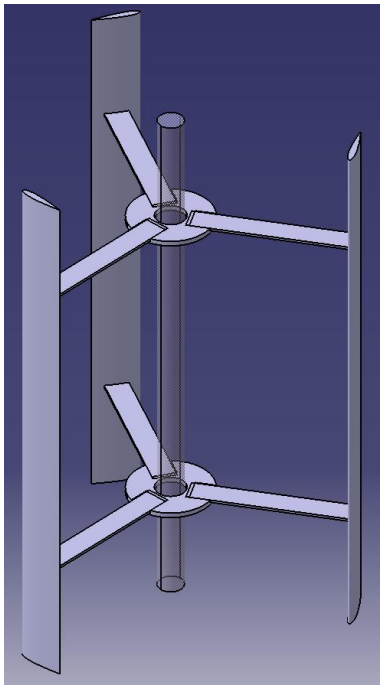


Figure 2.10 - Proposed straight bladed Darrieus design

Blade Height	2.07 m
Rotor Diameter	1.38 m
Chord Length	0.184 m
Solidity	0.4

Table 2.4 – Straight blade Darrieus design parameters

3. CFD Simulations and Analysis

3.1 [Simulation Setup]

Validating the simulation method is crucial for confirming the accuracy and reliability of the CFD results obtained using ANSYS fluent. 3D VAWT simulations have a high computational cost therefore, due to project time constraints, this report primarily uses 2D simulations. 2D analysis tends to underestimate tip losses and neglect strut effects which may lead to overprediction of torque, however, it provides a reasonable initial estimate of the turbines C_p-TSR curve. Samaouali et al [53] reported a max C_p of 0.4016 for 2D simulations and 0.5734 for 3D showing the 3D results deviating further from the experimental data.

3.1.1 [Geometry & Domain Definition]

To minimize errors from imported CAD geometries (e.g. from CATIA V5), the turbine and computational domain were made using native ANSYS tools (Design Modeler and SpaceClaim). Importing geometry via IGES or STEP can introduce topological errors, particularly in complex shapes such as VAWT blades [54], [55].

The computational domain is divided into two regions:

- Inner Rotating Region: Contains the turbine blades and has a diameter of 3R (where R is the rotor radius). The diameter of this zone prevents abrupt mesh size transitions which could cause numerical instability [56].
- Outer Stationary Region: Models the surrounding flow field. Although some studies (e.g. Rezaeiha [57]) recommend a domain size extending 10D upstream, downstream and laterally to minimize blockage effects, tests using a smaller domain used by Sagharichi et al. [34], shows only a 7% difference in C_p (see *Table 3.1*), with the larger domain having a 30% longer computational time.

Domain Test		
Size	Cp	% ΔCp
Large	0.2378	0
Small	0.2558	7.037

Table 3.1 - Effects of domain size on power coefficient

The chosen domain consists of an upstream distance of 7R, a downstream distance of 14R and top/bottom boundaries of 7R, this allows for a developed inflow profile and accurate wake capture. Zones for mesh refinement are also defined around the turbines wake and a 2c diameter circle around the blades to accurately capture effects of steep gradients in velocity, pressure and turbulence (domain is shown in *Figure 3.1*).

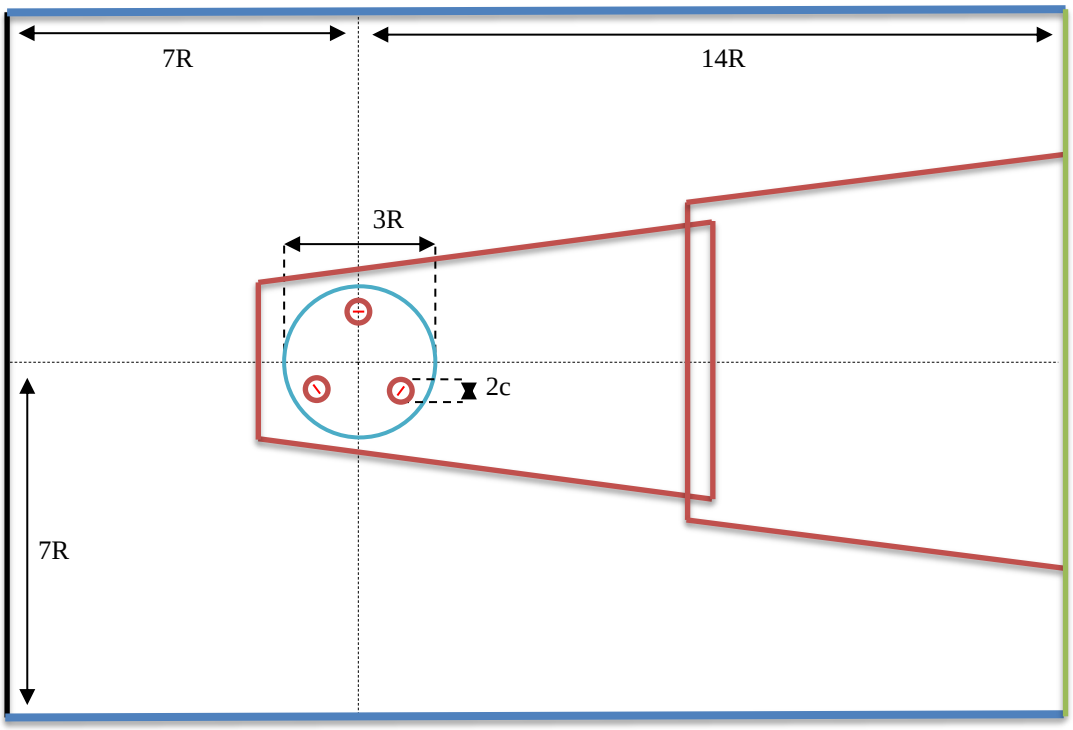


Figure 3.1 - Flow domain, Mesh refinement regions (Orange lines), Interface (Purple lines), Inlet (Black lines), Outlet (Green lines), Wall (Red lines)

The validation turbine is based on experiments by Castelli et al [58] (dimensions shown in *Table 3.2*) and is similar in size to the proposed designs which allows the methodology to be transferable. The NACA 0021 airfoil profile was taken from airfoiltools [59] using 200 points for accuracy. The turbine consists of 3 straight blades arranged symmetrically around the rotor center. The blades were shifted 0.25c to align with the blades aerodynamic center, which for symmetric airfoils such as the NACA 0021 is the point the aerodynamic moment is zero for all angles of attack, this is common practice in VAWT design and simulation [60].

Feature	Value
Rotor Diameter	1030 mm
Rotor Height	1 m (2D)
Number of blades	3
Blade Profile	NACA 0021
Chord Length	85.8 mm
Range of Tip Speed Ratio Tested	1.5, 2, 2.5, 3, 3.5
Free Strem Velocity	9 m/s

Table 3.2 - Validation turbine characteristics

3.1.2 [Mesh Generation]

Two mesh strategies were evaluated. Mesh 1 used a structured quadrilateral grid around the blades for consistent resolution in high gradient regions and mesh 2 uses inflation layers with triangular elements to capture near wall effects. For VAWT simulations y+ values of less than 1 are generally recommended [34], [61], [62].

For a given y+ the first layer height (y) can be determined, (derivation in Appendix B):

$$y = \frac{y^{+}\mu}{u_{\tau}\rho}$$

Equation 3.1

Both meshes have y+ of less than 1 around the blades to capture the boundary layer effects accurately. Conformal meshing was used to have similar cell sizes across all regions therefore, reducing interpolation errors. Mesh quality metrics followed ANSYS recommendations, with max skewness below 0.9 and average orthogonal quality above 0.95 [63], [64].

A mesh independence study showed that Mesh 1 had oscillating C_p values for the course, medium and fine meshes, likely due to abrupt cell size transitions, however, mesh 2 showed a more common convergence trend:

Mesh Independence (Mesh 2)		
Mesh Size	Cp	% ΔCp
Course (230,000 elements)	0.234	0
Medium (530,000 elements)	0.255	8.2
Fine (800,000 elements)	0.265	3.7

Table 3.3 - Mesh independence study for mesh 2

The medium mesh of 530,000 elements was selected as there is only a 3.7% change when moving to the fine mesh, without having excessive computational cost. *Figure 3.2* shows the torque vs azimuthal angle, for each three mesh sizes. This shows the expected 120 degree torque pattern repetition for a 3 straight-bladed VAWT.

In summary, the inflation layer based Mesh 2 offers smoother convergence and reliability making it the preferred choice for subsequent simulations.

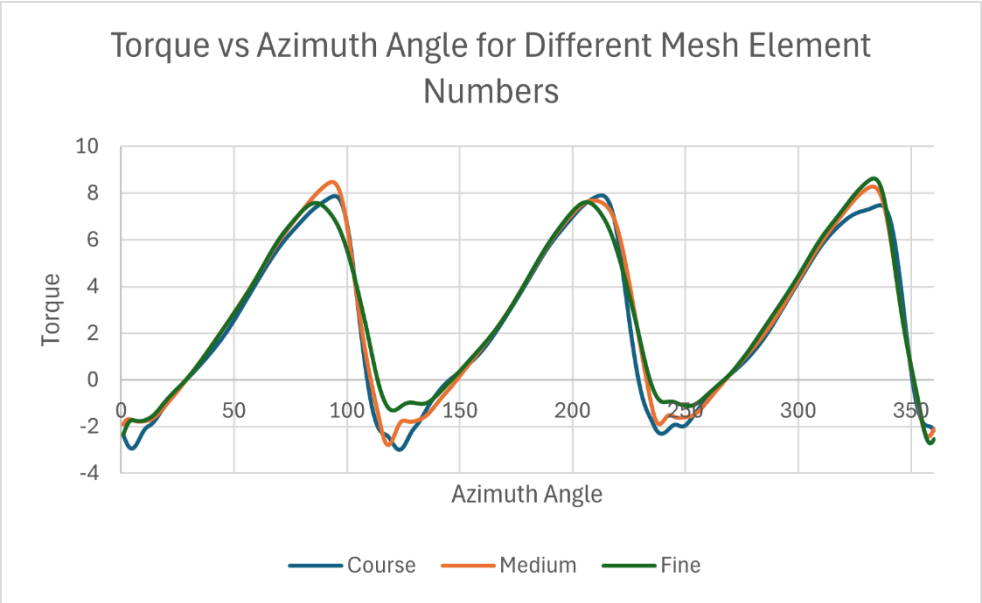


Figure 3.2 - Torque vs azimuthal angle for each mesh

Local blade meshes 1 and 2 are shown in *Figure 3.3*(a) and (b) respectively, *Figure 3.3* (c) shows the mesh of the outer to rotational domain.

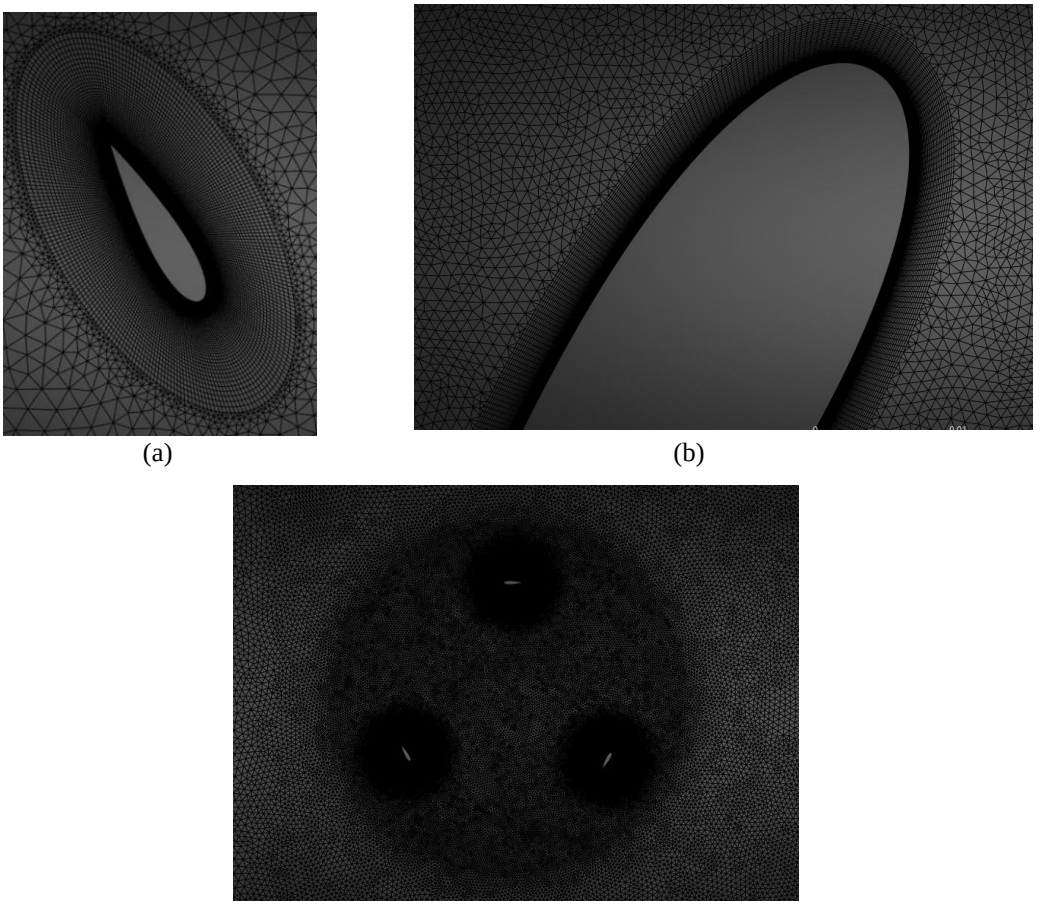


Figure 3.3 - (a) Mesh 1 local blade refinement (b) Mesh 2 local blade refinement mesh 3, (c) Mesh outer to rotating domain

3.1.3 [Domain Setup & Boundary Conditions]

With the simulation domain being divided into two regions, rotating and stationary domain an interface is required to enable smooth transfer of flow properties such as velocity, pressure, turbulence, etc. The rotating domain is then set to rotate at a specified tip speed ratio.

A timestep corresponding to a 1 degree azimuthal increment was chosen. Although Rezaeiha [57] recommends a minimum of 0.5 degree, upon review of their plot of of ΔCp against azimuthal increment, a 1 degree increment introduces only about a 2% change in C_p, therefore, a suitable balance between accuracy and computational efficiency.

Rotational velocity (ω) and timestep are calculated based on TSR (λ), turbine radius (R) and wind speed (U).

$$\omega = \frac{\lambda \mathrm{R}}{U}$$

Equation 3.2

Timestep (t) is calculated based on turbine circumference (C_r), tangential velocity (V_t) and azimuthal increment ($d\theta, \text{in degrees}$):

$$t = \frac{C_r}{V_t} * \frac{d\theta}{360} = \frac{2\pi R}{\omega \mathrm{R}} * \frac{d\theta}{360} = \frac{2\pi d\theta}{360\omega}$$

Equation 3.3

Table 3.4 lists ω and t for various TSRs for the validation turbine [58]. Residuals were set to the order of 10^{-5} for all criteria with 40 iterations per timestep. This allows for convergence and accurate capture of transient effects like vortex shedding.

TSR	Omega (rad/s)	Timestep (s) for 1 degree $d\theta$
1.5	26.214	0.0006658
2	34.951	0.0004993
2.5	43.689	0.0003994
3	52.427	0.0003329
3.5	61.165	0.0002853

Table 3.4 - Time step and rotational velocity for various TSRs

The inlet the wind velocity is set to 9 m/s with a turbulence intensity of 1% and turbulence length of 0.1c, these settings represent the controlled wind tunnel conditions the experiment would take place in, and are also used in the methodology of Hemiche Iddou et al [61]. The outlet is set to zero gauge pressure and the top and bottom boundaries are specified as zero shear to prevent unwanted boundary effects.

Convergence was of the solution typically happened by the 5th revolution (torque changes <1%), therefore, data from the 6th revolution onwards were used in future simulations (see *Figure 3.4* (a) and (b)).

The $k-\omega$ SST turbulence model was chosen for its efficiency in capturing near wall effects and its common use in VAWT simulations and is used with second order spatiotemporal discretization (*Table 3.5*). Literature shows that the C_p-TSR curve is often shifted in CFD simulations, reference [61] suggests using the PISO solver over the more commonly used coupled solver. Both coupled and PISO solvers were tested in which both solvers showed a rightward shift in the C_p-TSR curve. This trend is also reported in [65], [66] where they attribute the curves shift to the 3D effects and wake vortex blade interaction which can't be predicted in 2D analysis. The PISO solver followed the Cp-TSR curve slightly better than the coupled solver, however, with more overprediction in C_p (see *Figure 3.8*)

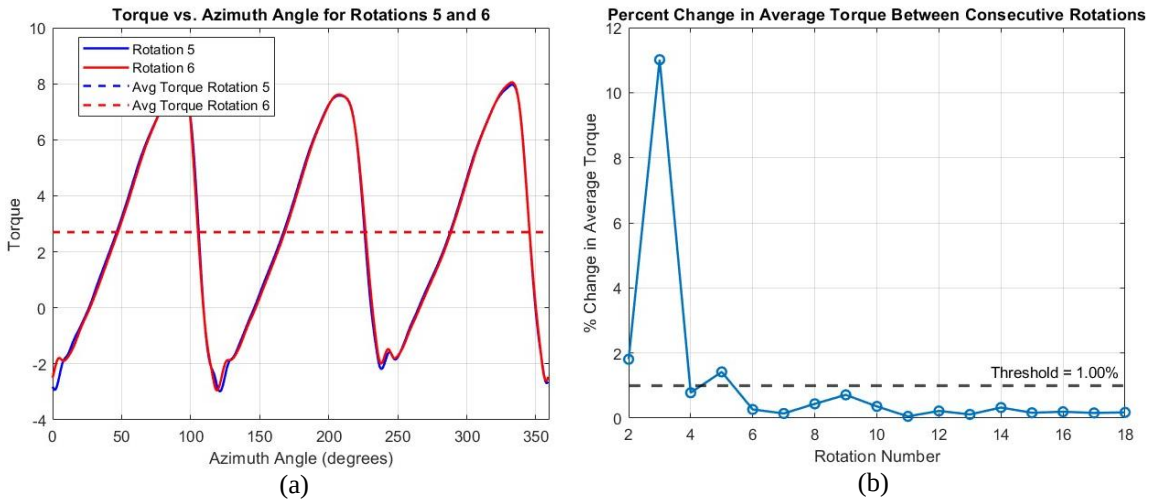


Figure 3.4 – (a) 5th and 6th turbine rotation torque vs azimuthal Angle, (b) Percentage change in torque vs rotation number

Gradient	Green-Gauss Cell Based
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
Transient Formulation	Second Order Implicit

Table 3.5 - Spatiotemporal discretization schemes

3.1.4 [Validation of Methodology]

The following contours and distributions from the CFD simulation are presented to validate that the numerical model accurately captures the aerodynamic behaviour of the VAWT.

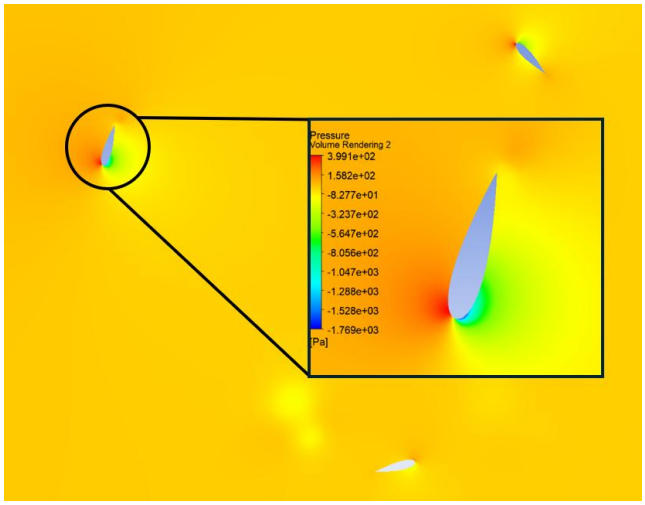


Figure 3.5 - Pressure distribution on the VAWT blades

The pressure distribution around the three Vertical Axis Wind Turbine blades, as shown in **Error! Reference source not found.**, shows expected aerodynamic characteristics. This shows both high and low-pressure regions along the leading edge of the blades. These pressure variations are likely caused by the relative motion of each blade with respect to the freestream wind and the wakes of the other blades. As each blade moves through different phases of its rotational cycle, it encounters varying pressure forces leading to asymmetric pressure distributions. Some regions of high pressure may be due to direct wind impact, while low pressure zones may be due to local flow acceleration or vortex interactions.

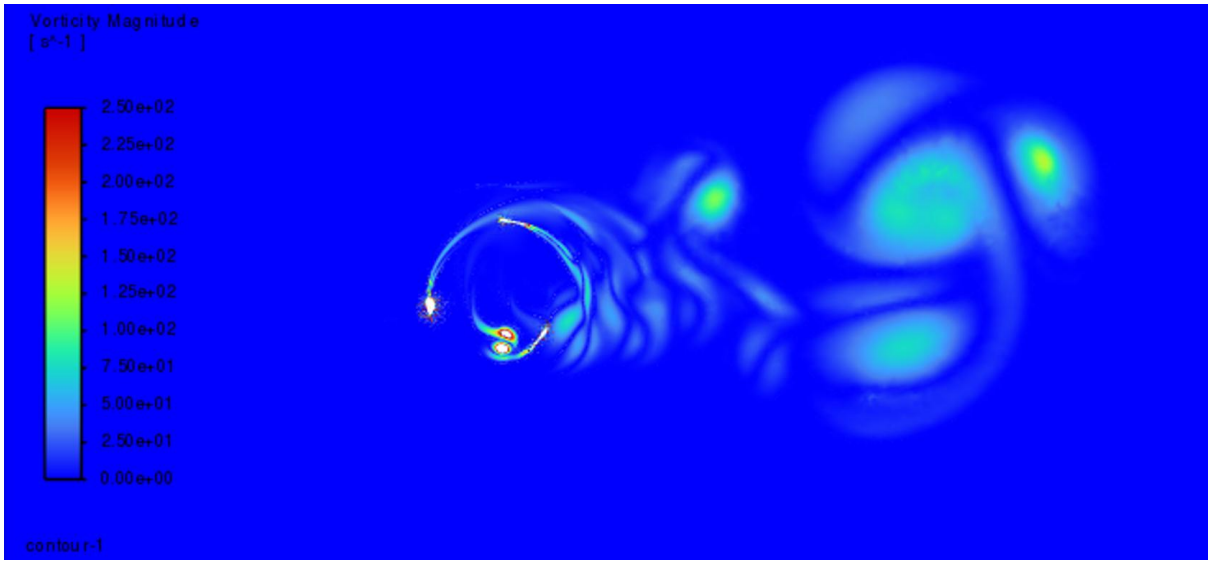


Figure 3.6 - Vorticity magnitude and wake development

Figure 3.6 shows the wake evolution and vortex shedding patterns before full convergence. The formation of vortex structures behind the blades indicates the expected unsteady aerodynamics associated with VAWTs. These vortices result from the periodic movement of the blades through the flow, generating alternating high-vorticity regions. The presence of a well-defined vortex structure suggests that the wake follows the characteristic pattern observed in experimental studies of VAWTs. However, some regions show strong localized vorticity near the blades, which could contribute to transient aerodynamic forces. This is why it is essential to make sure that the vorticity levels decrease in the final converged solution to prevent excessive turbulence-induced losses.

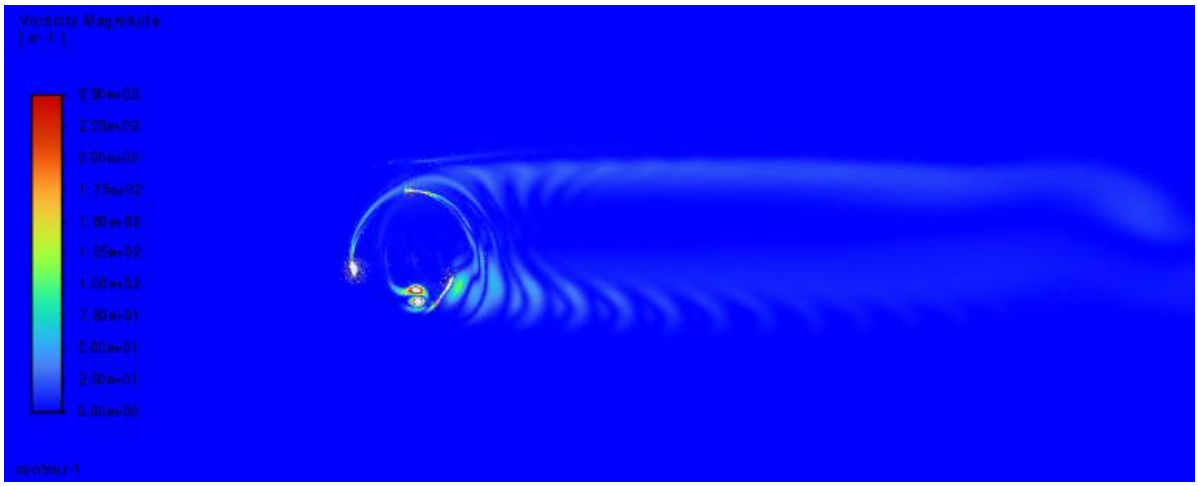


Figure 3.7 - Vorticity magnitude in the fully developed wake

The vorticity magnitude contour for the converged solution, shown in Figure 3.7, provides a detailed representation of the fully developed wake profile behind the VAWT. Compared to the pre-convergence vorticity distribution, this plot demonstrates a more stabilised and structured vortex shedding pattern, indicating that the wake has reached a steady-state condition. There is a clear reduction in vortex strength compared to the pre-convergence case, suggesting that transient effects have dissipated and that the simulation now represents a more time-averaged realistic flow. The highest concentrations of vorticity are located near the blade regions, aligning with areas of steep gradients observed in the pressure contour plot.

Figure 3.8 shows the simulation results against the experimental data and other computational results (compiled in [67]). The present work shifts the C_p towards the higher TSRs by about 0.5 TSR, this trend is consistent with literature and acceptable for predicting turbine efficiency and power output.

Comparing mesh strategies, the structured mesh (mesh 1) aligns more closely with the experimental data at lower TSRs but overpredicts at higher TSRs, mesh 2, when paired with the coupled solver shows around a 20% overprediction in C_p at optimal TSR which is expected in 2D analysis. Reference [47] showed losses due to struts can account for around a 26% loss in maximum efficiency, other inaccuracy could be attributed to inherent simulation uncertainties discussed previously.

Based on the mesh independence study and results shown in *Figure 3.8*, mesh 2 is selected for further analysis. The PISO solver with mesh 2 (blue line) captures the general trend slightly better, it overpredicts C_p by approximately 35%, therefore, the coupled solver is chosen due to its lower C_p overprediction and acceptable trend capture.

An alternative explanation for the rightward shift in the TSR is that real-world mechanical and electrical inefficiencies such as gearboxes, typically reduce the optimal TSR to about 60-75% of the theoretical prediction, which aligns with the CFD results obtained [68].

In summary, the present simulation methodology reproduces the C_p-TSR trend with around a 0.5 TSR shift which is comparable to other computational studies. This shift along with the overprediction in C_p will be considered in the subsequent analyses.

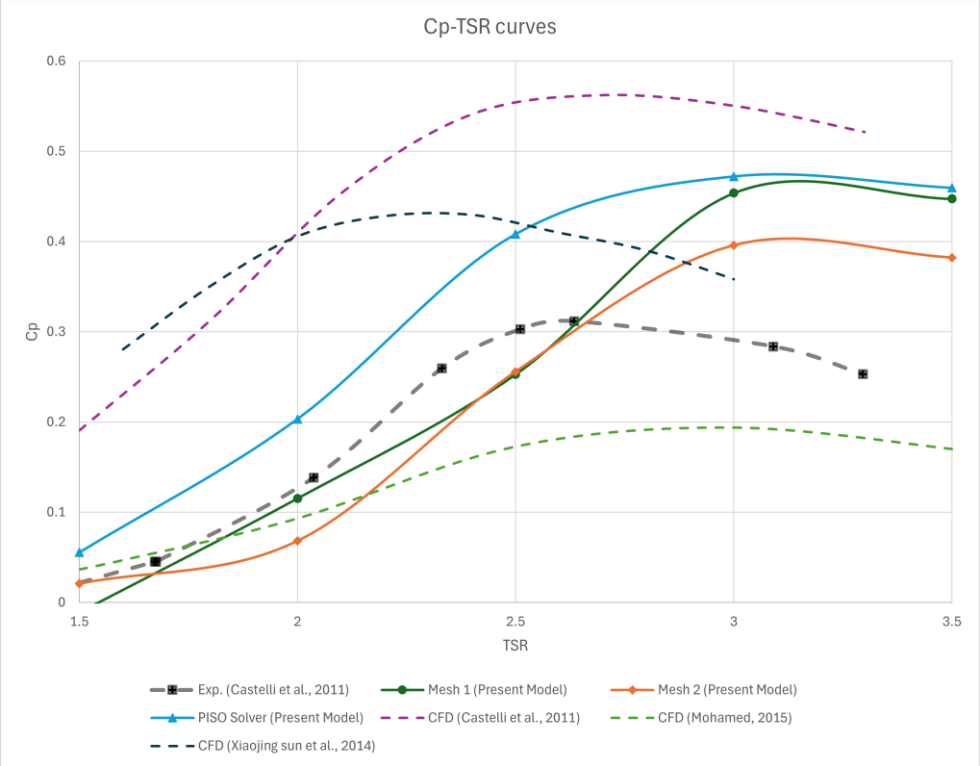


Figure 3.8 - Cp-TSR curves against experimental data, and other computational results and comparison of both mesh types and coupled vs PISO solvers.

Overall, the results validate the accuracy of the simulation and suggest that the numerical model effectively captures the aerodynamic performance of the VAWT.

3.2 [Analysis of Simulation Results]

The following sections will analyze the simulation results obtained from the proposed design geometry. Both the NACA0015 and 0018 blade profiles will be analyzed, the turbine will be simulated across a range of solidities, and ultimately a final design will be proposed based on the data gained from the CFD simulations.

For the 2D simulations, the inlet wind speed is set to 4 m/s (average site speed at turbine height), and the rotor dimensions match that of the initial straight-bladed Darrieus design, corrections for the helical turbine will be made based on empirical data.

It should also be noted that urban turbulence might increase drag and efficiency losses. Simulations were run with 1% turbulence intensity to ensure consistency with the validation study. However, future work should explore using higher turbulence intensities or test different under different conditions experimentally to capture real world urban wind behavior.

3.2.1 [Blade Profile Selection]

As explained in section 2.2.3, the blade profile is a design parameter which depends on the turbines optimal TSR, and is typically chosen to best fit the intended site. Both the NACA0015 and NACA0018 airfoils are recommended for different TSRs and wind conditions. After running simulations with both airfoils, *Figure 3.9* is obtained.

From the plot it can be observed that NACA0018 has a higher peak power coefficient at around 0.4 at a tip speed ratio near 2.6, in contrast to NACA0015 which peaks at roughly 0.38 with an optimal TSR of around 2.8. This means under optimal conditions NACA0018 should be able to extract more energy from the wind with the current turbine geometry. Although the 0.02 difference is small it can translate to notable increases in annual energy production over the turbines lifetime.

In the lower TSR range of around 1.5 - 2.5 (which is often where small VAWTs in unstable wind conditions would operate), NACA0018 also has consistently higher C_p than NACA0015. This would imply that across a wider range of wind speeds the turbine should have better performance on average with NACA0018, and suggests this airfoil is more tolerant to operation deviating from the design wind speed which is common in turbulent urban sites.

With the blades having a small chord lengths, the extra 3% thickness of the NACA0018 could bring quite a substantial increase to its structural rigidity, and should offer increased durability which is beneficial in urban environments with highly fluctuating wind loads. This could also extend turbine lifespan therefore improving ROI. FEA testing could be done to evaluate the extent of the structural performance increase over NACA0015.

As previously discussed in section 2.2.1, larger blade surface area increases interaction with the wind which can therefore improve starting torque. This could also give the thicker NACA0018 an advantage in self-starting ability, this is also seen in [69] which shows ‘much better self-starting characteristics’ using NACA0018 over NACA0012 .

NACA0015 does hold a small advantage in performance at higher TSRs, however, due to the low and variable site wind speed high TSRs will likely be less frequent. Higher TSRs can also increase wear and stresses on turbine components and lead to a shorter lifespan.

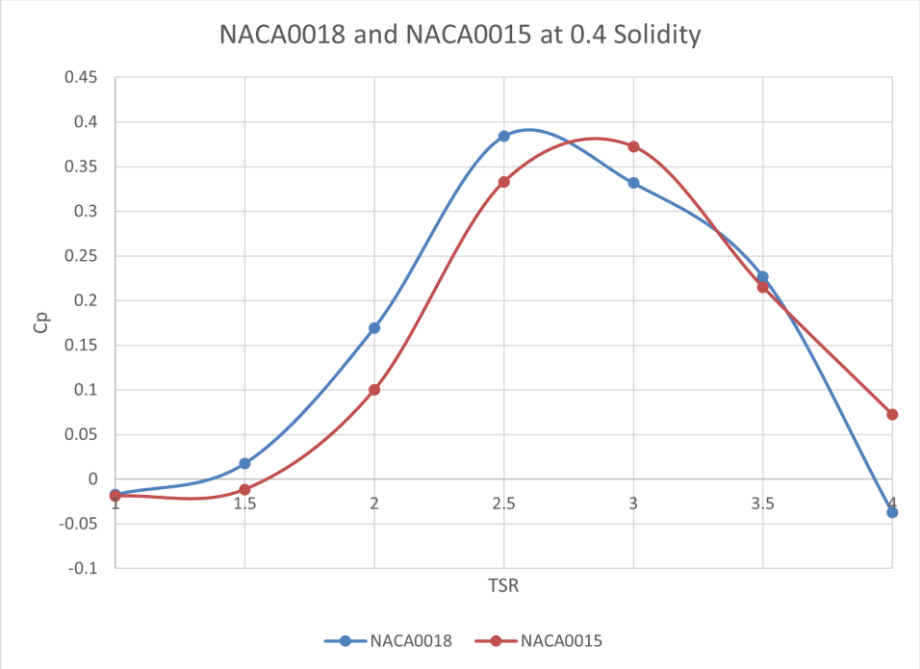


Figure 3.9 – NACA0018 vs NACA0015 C_p-TSR curves

Overall, considering the site and simulation results NACA0018 is a more suitable choice for the turbine, with higher durability, self-starting and a modest but consistent performance edge over the NACA0015 airfoil in the practical TSR range.

3.2.2 [Effect of Solidity on Performance]

As stated previously [39] and [50] recommend different solidities for turbines implemented at similar sites, varying from mid to high (0.3-0.5). To account for this, different solidities were tested to determine the most suitable turbine dimensions for the final design, based on the chosen semi-urban environment. NACA0018 airfoil was used based on previous analysis. After running the simulations across solidities: 0.2, 0.3, 0.4 and 0.5 the following C_p-TSR curves can be obtained (*Figure 3.10*).

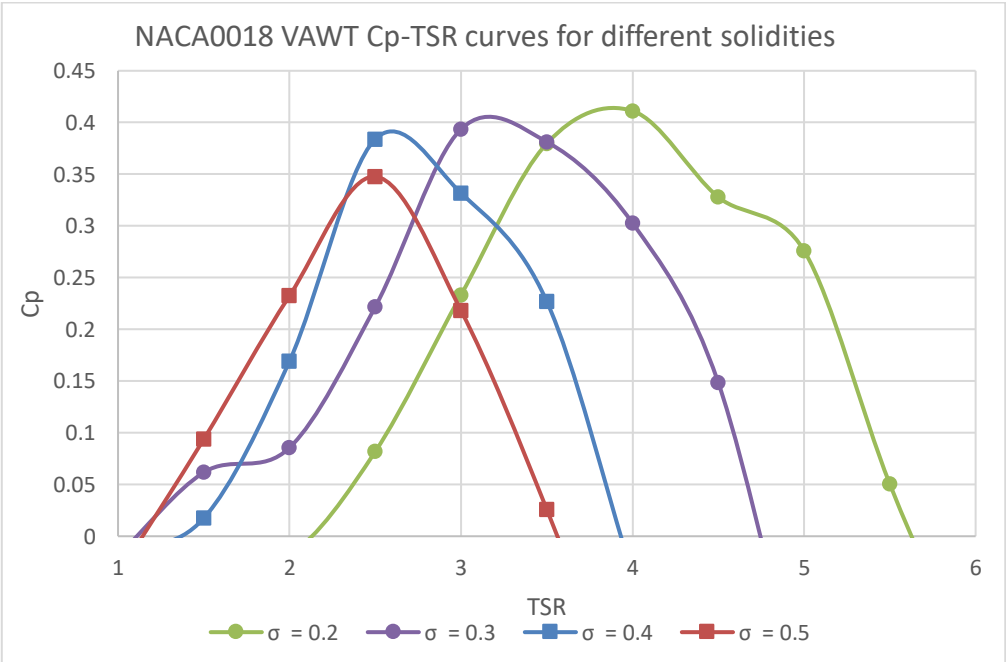


Figure 3.10 – Cp-TSR curves at different solidities

The results show the Cp increasing with decreasing solidity, this trend is consistent with literature and occurs due to the lower blade surface area (lower solidity, therefore, smaller chord length) decreasing drag. Due to the low wind speed this effect is not particularly prominent, as the larger blade surface area of higher solidity turbines can capture more energy from weak airflow.

The optimal TSR is also seen to shift into higher TSRs with decreasing solidity. This trend is also consistent with literature, and is due to less blade area extracting lower energy from the wind, the turbine therefore, must operate at higher TSRs to increase the pressure on the blades which helps to extract energy more efficiently. Lower TSR is often beneficial in sites with lower wind speeds as higher torque is generated at low speeds, which improves its startup ability and its efficiency in capturing energy even at low wind speeds.

Finally, the curve bandwidth widens with decreasing solidity, this trend is also seen in literature and occurs due to the higher solidity turbines producing higher drag and therefore, making turbine performance very sensitive to the TSR and only produce useful C_p over a narrow range of TSRs. Increased bandwidth means the turbine can operate more efficiently over a wider range of wind speeds with less severe drops in performance, and is more robust to fluctuations in different wind speeds. However, different trade-offs should be considered such as reduced performance at low TSRs and lower starting torque.

Each C_p-TSR curve was analysed in MATLAB using the Weibull distribution to find the AEP for each turbine. The AEP calculation followed the same method as the comparison code, assuming constant rotor speeds at different TSRs. At each wind speed the corresponding Cp value was determined from the Cp-TSR curve, and *Equation 2.1* was used to calculate power. This power was then summed over all hours in a year to obtain the total AEP. The C_p-TSR curves C_p values were reduced by 20% and the curve was shifted 0.5 TSR into the lower TSRs, to correct the predictions based on the CFD validation in section 3.1.4. The analysis showed interesting results with no clear correlation between AEP and solidity, the results are shown in *Table 3.6*.

Solidity	AEP (kWh)	Meets Monthly Requirement
0.2	155.6	Yes
0.3	176.3	Yes
0.4	138.1	No
0.5	184.3	Yes

Table 3.6 – AEP for different solidities

It can be observed that only the turbine with 0.4 solidity fails to power the streetlight. The AEP values don't follow a simple trend with solidity such as decreasing or increasing consistently, which may seem strange when looking at the curves shown in *Figure 3.10*, however, this behaviour can be explained when considering the assumptions made and how the turbine operates across the wind speed distribution. The assumption of the rotor speed remaining fixed (section 2.2.2), limits the turbine operation to the optimum TSR only at the average wind speed, when in actuality, the rotational speed may vary. This assumption is necessary as rotational speed at different wind speeds can't be determined without extensive dynamic mesh CFD simulations across different wind speeds, which could be carried out in further work. Therefore, this assumption actually emphasises the turbines performance across the entire range of wind speeds, which is particularly important in zones with highly variable wind speeds, such as more urban areas.

From the curves in *Figure 3.10*, it can be seen that for 0.4 solidity the curve drops off more severely in the lower TSRs and less in the higher TSRs. Based on *Equation 1.1*, increasing wind speed decreases TSR, therefore, due to a more severe C_p drop off in lower TSRs than other solidity turbines' it fails to efficiently convert higher wind speeds into energy. Whereas, the 0.5 solidity turbine has very consistent performance both below and above the optimal TSR, therefore, taking advantage of a wider range of wind speeds despite its lower C_p. The results also align with theoretical

assumptions in that higher solidity performs better at lower wind speeds and the recommendations for VAWTs in low wind speed areas, [50] recommends mid-range solidities, and from table 3.6, 0.3 solidity has the second highest AEP. [39] recommends solidities from 0.47 – 0.51, and table 3.6 shows 0.5 solidity with the highest AEP.

3.2.3 [Final Design Revisions]

Following the analysis of the turbine at different solidities final turbine dimensions can be determined. From *Equation 2.1*, the turbine power output increases proportionally with the swept area, this was then varied to find the smallest radius for each solidity turbine which could power the streetlight across all months. An additional 5% power production is also required to account for mechanical inefficiencies and abnormal wind conditions the Weibull distribution or the low turbulence intensity may not account for. The results are shown below:

Solidity	Smallest Radius (m)	Chord Length (m)	Blade Height (m)
0.2	0.69	0.092	2.07
0.3	0.66	0.132	1.98
0.4	0.72	0.192	2.16
0.5	0.65	0.217	1.95

Table 3.7 – Turbine sizes for different solidities

The initial design originally had a swept area of 2.85 m^2, however, the 0.5 solidity turbine could theoretically power the streetlight with swept area of 2.53 m^2. This is a roughly 12% decrease in swept area and is not only beneficial for the turbines spatial footprint, but also helps to lower manufacturing costs. The cut-in speed was kept constant at 3.5 m/s for each turbine, however, as previously discussed increasing solidity enhances self-starting ability, so this would likely give the 0.5 solidity turbine an even larger advantage over the others. Further work could include running dynamic mesh simulations for the turbine across a range of wind speeds to determine each solidity's cut-in speed, however, 3.5 m/s is a conservative estimate for a 0.5 solidity turbine.

Based on the analysis the final turbine design parameters are shown below:

Blade Height (m)	1.95
Chord Length (m)	0.217
Rotor Radius (m)	0.65
Blade Profile	NACA0018

Table 3.8 – Straight blade final design parameters

Helical Configuration

The 2D simulation predicts the performance of a straight bladed Darrieus, however, does not consider the 3D effects associated with the helical configuration. To give an estimate of the helical turbines dimensions and performance, trends were compared between helical angles shown in literature [18]. As helical angle increases, the optimal TSR increases, the C_p also increases at a helix angle of 60°, however, drops below the max C_p of the straight blade turbine at higher helix angles. The curves shape is also very similar to the straight blade turbine at helix angle of 60°. A helix angle of 60° is therefore chosen due to its superior efficiency and lower operational TSR range. Increasing helix angle 0 – 60° increases the max C_p by around 10% and shifts the curve by around 0.4 TSR into higher TSRs. Based on this an estimate of the helical turbine's C_p-TSR curve is shown in *Figure 3.11* (the CFD corrections have also been applied in plot, there are some slight errors in data interpolation). The turbine's AEP was also calculated and shown in table 3.9.

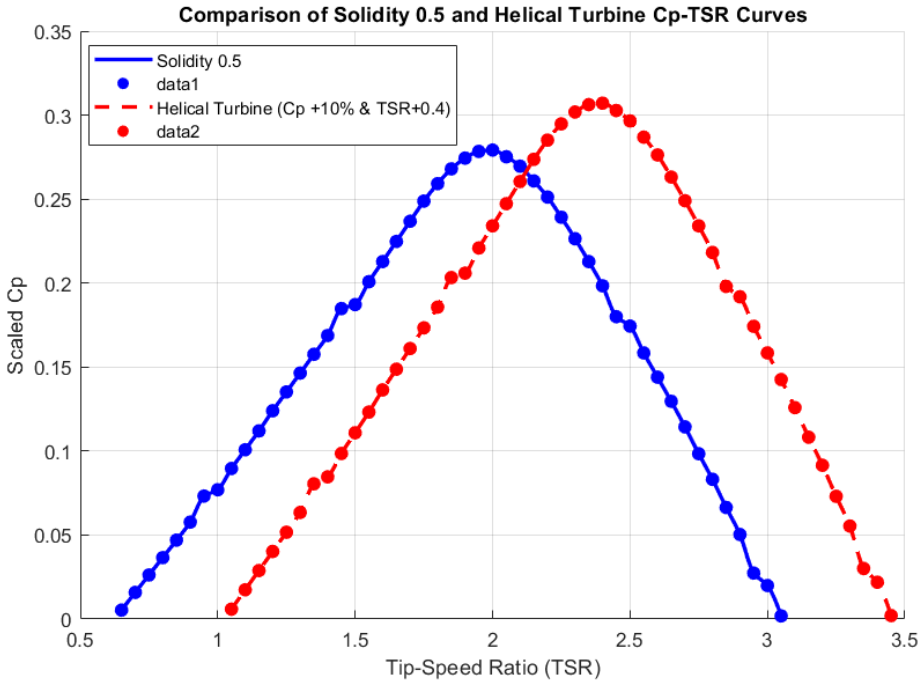


Figure 3.11 – Straight blade vs helical turbine C_p-TSR curve

Turbine	AEP at initial turbine dimensions (kWh)	Smallest radius (m)
Straight Bladed	184.3	0.65
Helical Turbine	161.5	0.67

Table 3.9 – AEP comparison for straight and helical turbines

Contrary to earlier predictions, the helical turbine actually has a lower AEP than the straight bladed Darrieus. The shift of the C_p-TSR curve increases the optimal TSR and makes the turbine spin faster aligning its peak efficiency with higher wind speeds. However, the wind speed distribution has many hours of low wind speeds in which the straight blade turbine would more effectively produce power, and the 10% additional C_p is not enough to offset this. This result aligns with recommendations in [50], which recommend a straight blade Darrieus for low wind speed sites. Despite this, the helical turbine is known to have lower cut-in speeds, lower noise and aesthetics as spoken on in section 2.3.1. It could still be considered, as there is only a 6% increase in turbine swept area.

It should be noted that the helical turbine AEP estimates will not be entirely accurate, as the turbine dimensions vary from those used in [18]. The cut-in speed of the helical turbine is also typically lower than that of a straight blade turbine. For example, from the sensitivity analysis conducted in section 2.2.1, decreasing the cut-in speed from 3.5 to 3 m/s could increase AEP by around 6% therefore, offsetting the higher 6% swept area. Future work could find the cut-in speed of each turbine and obtain a more accurate C_p-TSR curve, this would require 3D dynamic mesh simulations which are computationally infeasible for the scope of this project, however, the current 2D simulations serve as a good estimation of turbine performance.

The final helical turbine design is shown below:

Blade Height (m)	2.01
Chord Length (m)	0.224

Rotor Radius (m)	0.67
Blade Profile	NACA0018
Helical angle (°)	60

Table 3.10 – Helical turbine final design parameters

3.2.4 [Material Selection & Cost Metric Evaluation]

Material selection and installation costs are critical for the economic feasibility of small VAWTs designed solely to power streetlights. Conventional rooftop VAWTs can take centuries to break even for the average consumer [70]. However, if government funded and procured in bulk, costs, and therefore the return on investment (ROI) can be notably reduced.

For some small turbines, installation and production costs can reach around £3000 per unit, but these systems typically produce thousands of kWh annually. In contrast 30W streetlights consume around 120kWh/year. A turbine dedicated to streetlight power can therefore be smaller and less expensive [70], directly wiring the turbine to the streetlight also eliminates grid integration expenses (inverters, permits, utility fees), which can account for about the majority of costs for consumer installations [70].

Material Considerations and manufacturing costs:

Materials should be lightweight, strong, and corrosion-resistant. While aluminum provides a good strength-to-weight ratio, its high cost makes it impractical for small VAWTs. Composites like fiberglass or carbon fiber reinforced polymers offer good performance but may have high production costs. Carbon Glass Reinforced Epoxy (CGFRP-EP) strikes a balance between strength, durability, and cost. [71].

VAWTs typically last 20-25 years [19]. CGFRP-EP resists UV radiation, moisture, and corrosion, making it well-suited for outdoor applications. It also offers superior tensile strength and fatigue resistance compared to plastics, reducing the risk of cracking under cyclic loads. Since this design operates at lower wind speeds (4 m/s vs. 8+ m/s for larger VAWTs), mechanical stress is lower, supporting a 25 year lifespan.

For the straight-bladed Darrieus design, simpler manufacturing methods (extrusion or laser cutting) can lower costs. The turbine consists of three blades, each measuring 1.95 m in length with a chord of 0.217 m. Assuming a simple rectangular profile, each blade has an approximate surface area of:

$$Area \approx 2sides * 1.95m * 0.217m \approx 0.846m^2$$

Thus, the total surface area for all three blades is approximately 2.54 m^2.

Blades are made from a 50% carbon, 50% glass hybrid fabric (180 g/m^2). Since a single layer is too thin and may crack under wind loads, two layers per side are recommended, doubling wall thickness and improving durability.

Total fabric required:

$$2.54\ m^2 * 2sides * 2layers \approx 10m^2$$

Based on sliding scale pricing [72], 10m^2 costs £93, (plus 10% waste), fabric costs about £102.

With a 50% fiber volume fraction fabric weight (0.18 kg/m^2) totals 1.8 kg. To balance the composite, resin weight must also be 1.8 kg. Given a 2:1 resin-to-hardener ratio, total resin kit weight is:

$$Total\ kit\ weight = \frac{1.8\ kg}{(\frac{2}{3})} \approx 2.7\ kg$$

Including 10% waste, 3 kg of resin kit costs about £30. [73].

Additional consumables (release agents, cups, gloves, brushes) add £10, bringing the total blade material cost to ~£142.

Structural and mechanical components:

Component	Specification	Cost per Unit (£)	Quantity	Total Cost (£)
Bearings	Ball bearings to allow shaft rotation	10	2	20
Generator	Small DC motor (1 kW, aligns with Betz limit at max wind speed, [74])	40	1	40
Mounting Hardware	Bolts, nuts, washers (bulk order)	-	-	10
Shaft	2 m steel pole, [75]	7	1	7
Casing	2 m x 1.4 m case (from recycled steel, estimated)	30	1	30
Breaking system and wiring components	Small brake pads and inexpensive wires	15	-	15
Total Cost				£122

Figure 3.12 - Structural and mechanical components

Total Material Cost: £264 per turbine

Manufacturing and Labor costs:

Assuming an hourly labour rate of £15:

- Blade Fabrication (£75 total): Mold prep (£15), Fabric cutting & layup (£30), Curing & demoulding (£15), Trimming & finishing (£15).
- Assembly & Installation (£60 total): Shaft prep (£15), Bearings & mounting (£15), Generator integration (£15), Final assembly & testing (£15).
- Total Labor Cost: £135

Final Estimated Cost per Turbine:

$$Total\ Cost = \ Materials\ (£264) + Labor\ (£135) = £399$$

Installation and Maintenance costs:

Mounting the turbine directly on a streetlight eliminates grid connection expenses. Installation costs are estimated at £300 per unit for a single turbine but drop to £100 per unit for bulk installations of 20+ turbines (20 units along a 906 m street like Baldock Road, in the zone used to get the average site wind speed [76], [77]), due to shared mobile elevating work platform (MEWP) hire.

- Single Unit Installation: ~£699 total cost per turbine
- Bulk Installation (20 units): ~£433 per turbine (includes 25% material cost reduction from bulk orders, calculated by sliding prices (100 * (€8.45 - €11.1) / €11.1$\approx 24\%$) [72]).

Maintenance is estimated based on a repair time of 20 minutes per turbine, with an assumed labour rate of £15/hour. This results in a labour cost of £5 per turbine per year. Parts replacement is estimated at £5 per turbine annually.

For individual maintenance, could cost ~£315 per year per turbine (£15 in labour + 2 yearly trips at £150 each for MEWP hire) [78]. However, in bulk maintenance scenarios (e.g., 20 turbines), MEWP hire costs can be distributed across turbined, reducing per-turbine maintenance costs to ~£30 annually.

Payback Analysis:

Assuming 125 kWh/year production at an electricity rate of £0.33/kWh, includes standing charge, government climate change levy, VAT and kWh cost [79], [80],:

- Annual savings per turbine: 125 kWh × £0.33 = £41.25.
- Single Installation Payback Period:
    - £699 / (£41.25 - £315) ≈ Never (maintenance costs exceed savings)
- Bulk Installation Payback Period (20+ turbines):
    - £433 / (£41.25 - £30) ≈ 38.5 years (exceeds lifespan)
- City-Wide Scaling (267 streets, 5000+ streetlights):
    - Bulk maintenance reduces per-turbine costs to ~£19.76/year (£15 in labour + 2 yearly trips at £250 MEWP hire, shared among 105 turbines) [78].
    - Payback period: ~20.14 years.
    - Potential savings over 25 years: (£41.25/yr - £19.76/yr) x (25yr - 20.14yr) x (267streets * 20 turbines) ≈ £560,000

The helical turbine design may have 15-30% lower savings in comparison to the straight bladed Darrieus due to increased manufacturing cost, but may be preferred due to its more compact size and reduced noise. The analysis suggests that small-scale or individual turbine installations are not economically viable due to high maintenance costs. However, large-scale, bulk installations could break even within the turbine's lifespan and generate substantial long-term savings. These estimates are based on generalized data, and site-specific studies are necessary to confirm feasibility.

3.2.5 [Environmental Benefits: Quantifying Emission Savings]

One of the key benefits of using VAWTs for streetlights is the potential for significant emission reductions. The annual energy requirement for a 30W streetlight is approximately 125 kWh. The average carbon intensity of grid electricity in the UK is 225 grams of CO_2 per kWh, reflecting a mix of renewable and fossil fuel sources [81]. However, in scenarios where electricity is primarily generated from fossil fuels, the carbon intensity can be much higher—ranging from approximately 465 g CO_2/kWh for natural gas to 980 g CO_2/kWh for coal. Based on the average grid intensity, powering the streetlight with grid electricity results in annual emissions of 125 kWh × 225 g/kWh = 28.125 kg CO_2.

In contrast, wind energy has a significantly lower lifecycle carbon intensity of around 11 grams of CO_2 per kWh [82]. If the same streetlight were powered by wind energy, the associated emissions would drop to 125 kWh × 11 g/kWh = 1.375 kg CO_2, resulting in a net reduction of 26.75 kg CO_2 per streetlight per year. This reduction would be even more pronounced if the alternative was a fossil-fuel-dominated grid, further emphasizing the environmental benefits of integrating small-scale VAWTs into urban infrastructure.

To show the scale for a semi-urban area with 100 such streetlights. The total annual CO_2 savings would be 27 kg * 100 = 2,700 kg, or 2.7 metric tons. This is a small but notable reduction, equivalent

to the annual emissions from about 2 average cars in the UK, each emitting around 1.4 metric tons yearly [83]. Scaling up, if a million LED streetlights in the UK were similarly powered, the total savings would reach 27,000 metric tons annually. In the context of the UK's national emissions, 371 million tonnes of CO_2 [84]. The savings from all streetlights, at 0.027 Mt (27,000 metric tons), make up approximately 0.007% of total emissions (0.027 / 371 ≈ 0.000073). This percentage is small, however, could show significant cumulative reductions over many years.

4. Project and Quality Management

4.1 [Project Management]

The initial Gantt chart provided a structural plan for the project, outlining key tasks and milestones. However, as the project progressed, some adjustments were necessary due to unforeseen challenges and refinements in the methodology. The revised Gantt chart reflects the reflects these changes, particularly in task durations and introduction of additional phases.

Critical activities in the project included data collection, wind speed analysis, MATLAB code development, and ANSYS simulations. These tasks were crucial because delays in any of them could significantly delay subsequent phases. One major difference between the initial and revised Gantt chart was the extension of the meshing and simulation phases. Originally, the plan did not account for the learning curve associated with ANSYS Fluent, nor did it anticipate the computational demands of 3D simulations.

Initially, time estimates for these tasks were overly optimistic. This was largely due to a lack of research into typical computational times for 3D VAWT simulations, which led to a later switch to 2D simulations. As the project progressed, it became clear that achieving accurate results required finer meshing and multiple solver adjustments, which significantly increased computational time. To keep the project on track, priorities were adjusted, and workflows were changed to allow for completion within the overall timeframe.

Key Changes and Their Impact

One primary change was the extension of the meshing phase due to the time required to learn ANSYS Fluent and refine mesh settings for accurate results. The initial plan underestimated this complexity, leading to inefficiencies in early simulations. Additionally, 3D simulations were abandoned in favour of refined 2D simulations to maintain feasibility.

Another major change was the introduction of a validation phase, which was not originally planned. As I progressed, I recognised the importance of aligning simulation results with experimental data. Testing different solvers, domain sizes, and mesh configurations extended the timeline but significantly improved result reliability.

These changes led to the exclusion of testing the turbine across multiple wind speeds due to time constraints. To compensate, report writing was started earlier while simulations were running. Despite timeline extensions, the project structure remained consistent with refinements in removal of tasks rather than fundamental changes.

Strengths, Weaknesses, and Improvements

A key strength of the project management was adaptability, adjusting workflows in response to challenges, therefore, allowing for progress without compromising the quality of the project. Efficient time utilisation such as doing the report in parallel with the simulation also helped in mitigating delays.

The main weakness was underestimating the learning curve for ANSYS Fluent and time required for accurate simulations. More thorough pre-project research on computational demands could have led to more realistic estimates. Additionally, planning to do validation from the start would have streamlined the process.

For future projects, I would allocate more time for learning complex software and refining simulation strategies before committing to a specific approach. More research into the timeframes for each task would also be done in the initial planning phase. Despite the challenges, the ability to adapt and optimise workflows allowed for the project's completion.

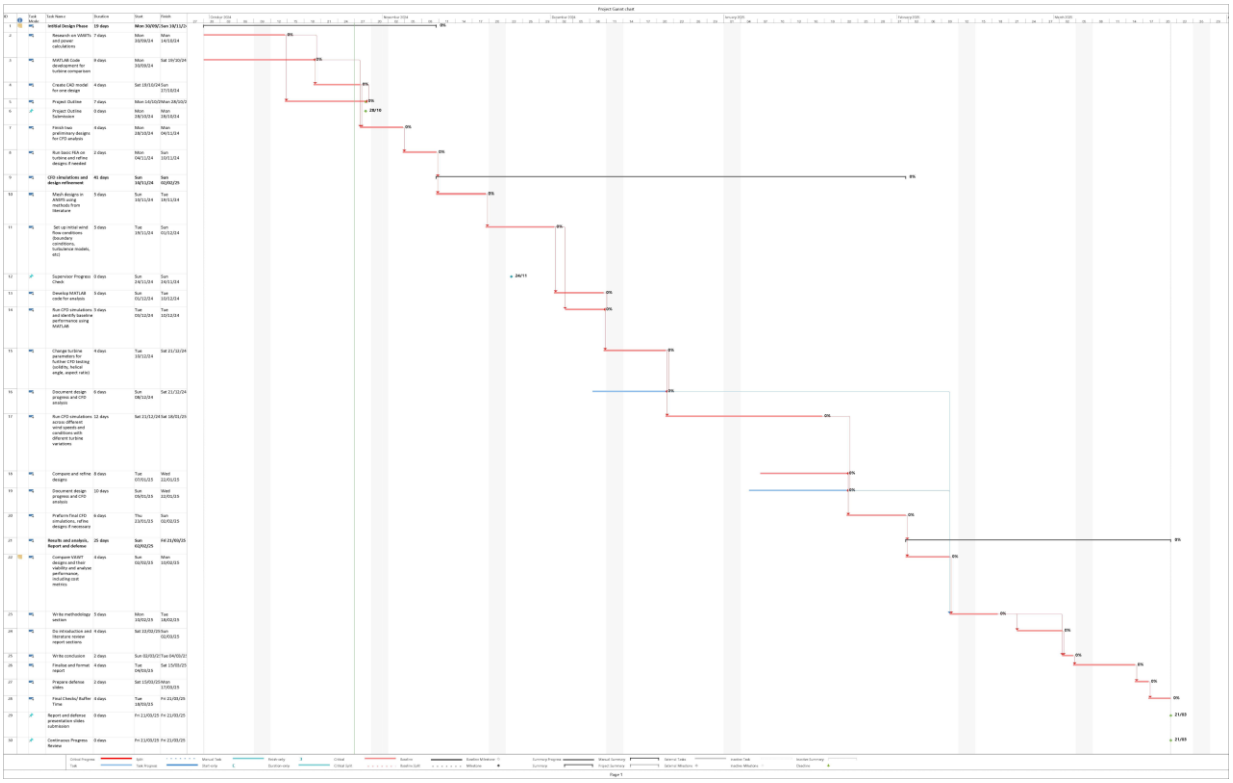


Figure 4.1 – Initial Gantt Chart

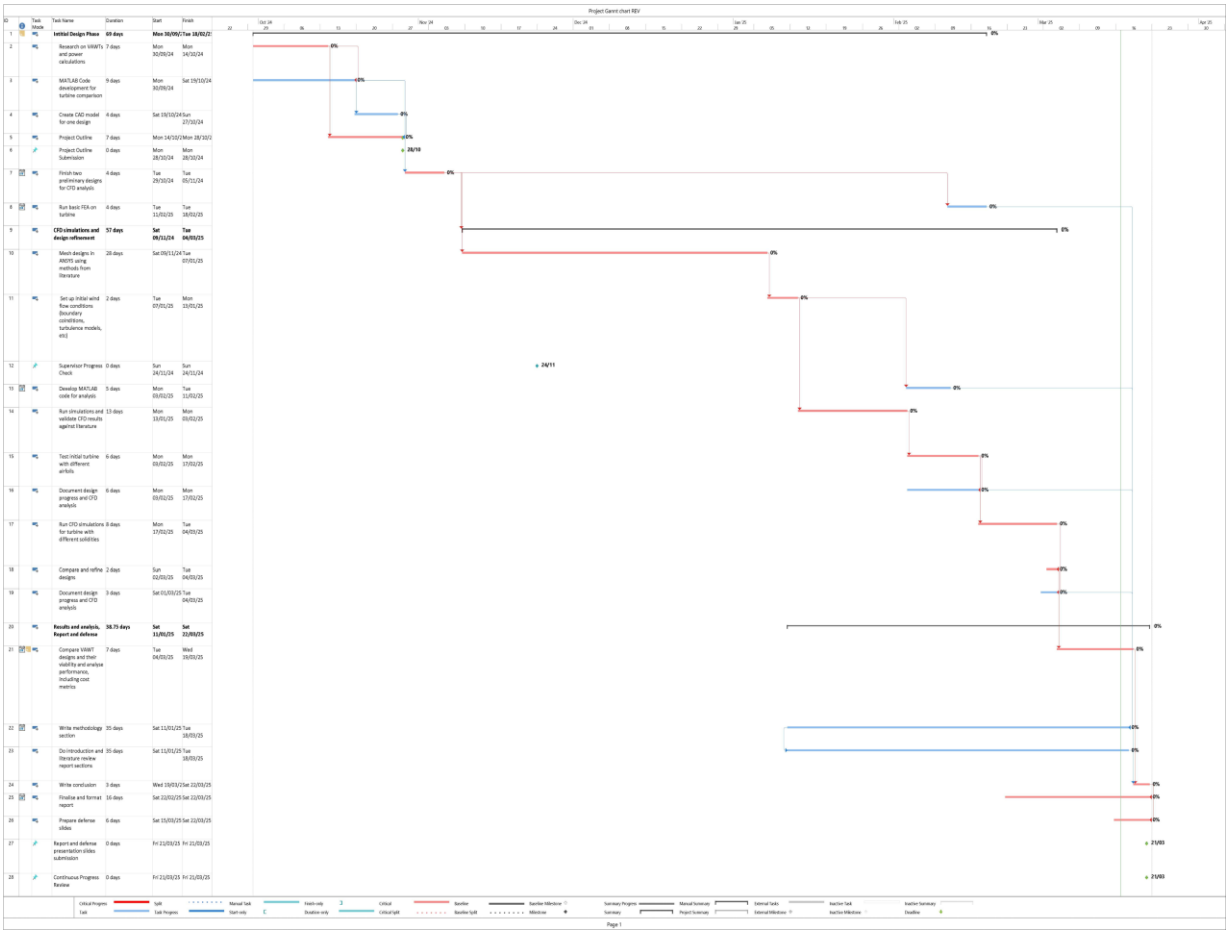


Figure 4.2 – Revised Gantt Chart

4.2 [Quality Management]

Standards in engineering projects ensure consistency, reliability, and accuracy, particularly in computational analysis. In this project, standardising simulation setup, meshing quality, and solver selection was essential for producing repeatable and meaningful results.

In the CFD simulations, maintaining consistency in turbulence models, boundary conditions, and mesh refinement was crucial to achieving accurate simulations. Following established practices, such as refining mesh until results stabilised and validating outputs against experimental data, helped improve accuracy and reliability. These practices make sure that findings are not only valid for this study but also useful for future research and potential real-world applications.

Strict adherence to best practices can be time consuming and computationally expensive. For example, refining mesh to improve accuracy significantly reduced simulation times. Validation efforts required testing different solver settings and domain sizes, which extended the project timeline. These challenges added complexities, however, were necessary trade-offs to ensure reliable results. Overall, the benefits of maintaining quality outweigh the drawbacks of increased computational time and added complexity.

One key area for improvement is the time management in quality control. Future projects would allocate dedicated time for validation rather than integrating them later in the process. Optimising and exploring different mesh refinement strategies could reduce simulation time and without compromising quality.

5. Conclusion

This project successfully designed and analysed a small-scale vertical axis wind turbine for urban streetlight applications, providing a sustainable alternative to grid-powered lighting. Through a combination of computational modelling, numerical scaling, and CFD simulations, the study developed an optimised turbine design suited for low and variable wind conditions.

The MATLAB framework provided accurate initial estimates of turbine geometry and power output, enabling comparisons between different VAWT configurations. 2D transient CFD simulations indicated that the straight-bladed Darrieus turbine outperformed the helical design, which required a 6% larger swept area to meet annual streetlight power demands. However, the lack of 3D dynamic mesh simulations make the full advantages of the helical turbine uncertain in low-wind conditions. The NACA0018 blade profile with a turbine solidity of 0.5 showed the highest annual energy production (AEP), with a recommended minimum swept area of 2.53 m² for powering a 30W streetlight annually.

A structural and cost analysis confirmed the feasibility of mounting these turbines on streetlights, assuming periodic maintenance. The estimated material and labour cost per turbine is approximately £399, with single-unit installations reaching £699. Bulk installations (20 turbines per street) could lower costs to £433 per unit. Payback analysis suggests that unless installed on a large scale (105+ turbines), the maintenance costs outweigh the initial investment. However, at large scales, the system becomes economically viable within its expected lifespan.

Shifting from grid power (225 g CO_2/kWh) to wind energy significantly reduces emissions, with a single wind-powered streetlight emitting just 1.4 kg CO_2 annually compared to 28 kg from the grid. When scaled to one million streetlights, this transition could save approximately 27,000 metric tons of CO_2 per year. However, this reduction represents only 0.007% of the UK's total emissions, suggesting that while beneficial, widespread adoption would be driven more by economic savings than carbon savings alone.

Integrating these turbines at a large scale, such as across 267 streets with approximately 20 streetlights each, could see savings of around £560,000 over a 25 year turbine lifespan. This demonstrates the small-scale VAWT as both an economical and environmentally viable option for reducing carbon footprints and electrical costs.

Despite these results, some limitations remain:

- CFD results have some inaccuracies, likely due to 2D modelling limitations and neglecting 3D wake interactions.
- Lack of experimental validation, meaning real-world factors such as turbulence, wake effects, and mechanical inefficiencies were not fully accounted for.
- AEP predictions are not entirely accurate due to the assumption of a constant rotor speed.
- Cut-in speed of each turbine design remains uncertain.

Further work should focus on:

1. High fidelity 3D CFD simulations to more accurately capture flow physics and validate the C_p-TSR trends observed in 2D, and to determine rotor speeds across varying wind speeds and turbine cut-in speeds to get a more accurate prediction of each turbines AEP.
2. Wind tunnel testing to experimentally validate power output and cut-in speeds.
3. Structural load analysis in urban environments to evaluate long term operational reliability.

Overall, this project demonstrates that small-scale VAWTs offer a viable and sustainable energy solution for urban lighting applications, reducing reliance on traditional power grids and potentially saving thousands in electrical costs.

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APPENDIX A – MATLAB Scaling Methodology

To capture the effects of solidity, the MATLAB code implements a scaling methodology based on empirical C_p-σ-TSR relationships defined by Vishnu Namboodiri [85], this relationship is shown in **Error! Reference source not found.**.

Max Power Coefficient	Solidity	TSR_{opt}
0.37	0.09	4.5
0.40	0.12	4
0.43	0.18	3.5
0.44	0.24	3
0.45	0.30	2.5
0.45	0.36	2

Table 0.1 - Relationship between solidity and power coefficient for small VAWTs

As solidity increases, the optimal TSR decreases and C_p improves, however, plateaus around $\sigma = 0.36$. Solidities exceeding 0.6, however, may start to see lower C_p [86].

The solidity scaling process implemented in the MATLAB code follows the following process:

1. Empirical Baseline: Interpolating C_p-σ-TSR curves from experimental data.
2. Optimality-Preserving: Maintains the turbines deviation from its optimal solidity as its optimal TSR changes.
3. C_p Adjustment: Predicts the new TSR from design solidity, then applies a deviation factor to scale C_p accordingly

This method ensures, for example a turbine designed at 20% over the optimal solidity for TSR=4.5 scales to 20% over the optimal solidity at TSR=2.0.

Validation:

Validating the scaling methodology is performed by comparing general trends predicted by the code to experimental expectations shown in literature. More in depth validation could be performed if experimental studies of the same turbine at different solidities (due to change in chord length) whilst reporting their corresponding Reynolds numbers were carried out.

Predicting a turbines optimal TSR typically requires use of more computationally expensive methods such as CFD or BEM, however, the code estimates the optimal TSR based on the empirical relationship shown in **Error! Reference source not found.**. Figure 0.1**Error! Reference source not found.** shows the scaling for two turbines; helical and straight-bladed Darrieus. Originally the helical turbine had peak efficiency of 40% and the Darrieus 32%, however, after Reynolds and solidity scaling (adjusting both to 0.4 solidity and Re corresponding to site wind speed), the helical turbines peak efficiency dropped to around 15% below the straight blade design whilst having a broader operational range. These trends are consistent with what is reported in literature with lower peak Cp but a better performance range

The predicted optimal TSR of approximately 1.7 also aligns well with values reported for similar solidity VAWTs. For example, a similar straight bladed Darrieus turbine with 0.4 solidity was experimentally tested by Bravo et al. [87] which is a commonly used experimental study for CFD validation [61], [65], shows an optimal TSR near 1.6 and show similar Cp-TSR trends. Although the experimental data indicates a maximum C_p of 0.26 at 6 m/s wind speed suggesting that the scaled turbines maximum C_p is somewhat overpredicted at an average site wind speed of 4 m/s, this uncertainty is accounted for in the code.

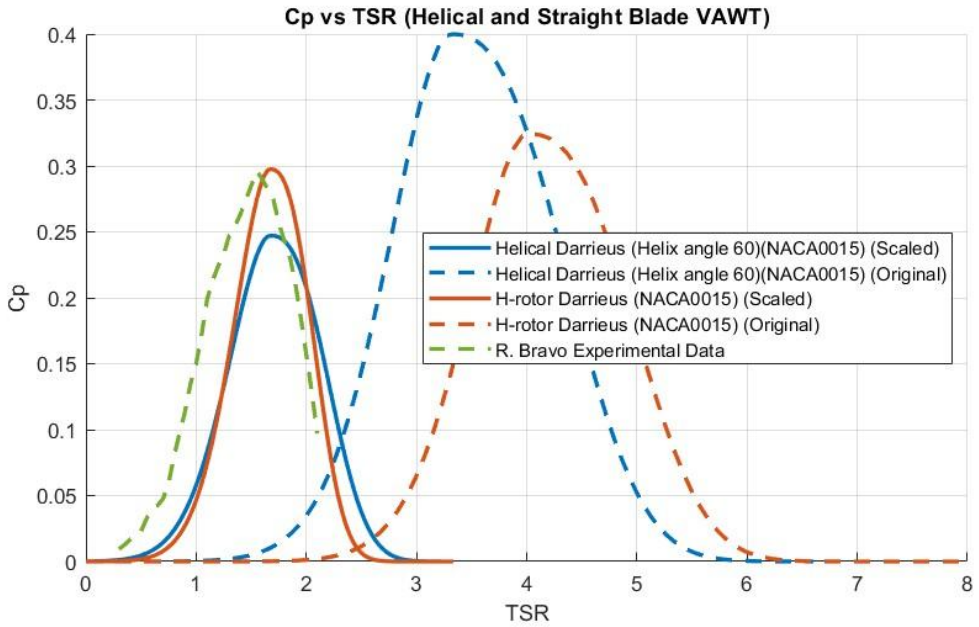


Figure 0.1 - *Comparison of two reference turbines before and after solidity and Reynolds scaling*

Overall, these results provide confidence that the scaling methodology can reasonably predict the C_p-TSR trends of VAWTs with low computational expense, though further refinement should be done for higher TSR turbine designs.

APPENDIX B – Extended Mathematical Derivations

Weibull Distribution:

The two parameter Weibull distribution is commonly used for wind speed predictions. The Weibull distribution function is shown in Equation 0.1. The parameter 'C' represents the scale parameter which determines the abscissa scale of the wind speed distribution. The parameter 'k' represents the shape parameter which reflects the width of the distribution. [88]

$$f(v) = \left(\frac{k}{C}\right)\left(\frac{v}{C}\right)^{k-1} \exp\left[-\left(\frac{v}{C}\right)^{k}\right] \ (v > 0; k, C > 0)$$

Equation 0.1

The scale and shape parameters can be estimated by using:

$$k = \left(\frac{\sigma_v}{V}\right)^{-1.086} \ (1 \leq k \leq 10)$$

Equation 0.2

$$C = \frac{V}{\Gamma(1 + \frac{1}{k})}$$

Equation 0.3

Where Γ is the gamma function, σ_v is the standard deviation of wind speed and V being the mean wind speed.

First Layer Thickness:

The first layer thickness can be estimated based on flat-plate boundary layer theory [89]:

$$Re_x = \frac{\rho U c}{\mu}$$

Equation 0.4

Reynolds number calculation using wind speed (U), chord length (c), dynamic viscosity (μ) and density (ρ). Skin friction coefficient (C_f) can then be calculated:

$$C_f = \frac{0.664}{\sqrt{Re_x}}$$

Equation 0.5

Wall shear stress (τ_w) can then be calculated, and then wall shear velocity (u_τ):

$$\tau_w = \frac{1}{2} C_f \rho u^2$$

Equation 0.6

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}}$$

Equation 0.7

Finally, for a given y+ the first layer height (y) can be determined:

$$y = \frac{y^{+}\mu}{u_{\tau}\rho}$$

Equation 0.8

APPENDIX C – MATLAB Cp-TSR approximation

To estimate the annual energy production of a VAWT, a C_p-TSR curve is approximated to model the turbine operating away from its maximum C_p. The code implements this by taking maximum Cp values, solidity, and optimal TSR from different turbine geometries in literature [18], [67], [90]. These allow for the approximation of a C_p vs TSR curve. Common urban turbine designs (helical, straight-bladed-Darrieus and Savonius) are compared with the focus on differences in general turbine geometry, rather than user-defined parameters like solidity.

Experimental data shows that Cp tends to drop quadratically around the optimal TSR. At low TSR, high AoA causes dynamic stall and flow separation, whereas at high TSRs, reduced AoA limits lift and drag, thereby reducing efficiency [91]. To capture this trend, two exponential decay functions are created:

Below TSR_{opt}:

$$C_p(TSR) = C_{Pmax} * \exp\left(-\frac{3.2}{TSR_{opt}} * \left(\frac{TSR_{opt} - TSR}{1.2 * \sqrt{Width\ Factor}}\right)^c\right)$$

Equation 0.1

Above TSR_{opt}:

$$C_p(TSR) = C_{Pmax} * \exp\left(-a * \left(\frac{TSR - TSR_{opt}}{WidthFactor * TSR_{opt}}\right)^b\right)$$

Equation 0.2

Constants a, b and c are tuned using data from literature and control the decay rate and steepness of the curve. The width factor controls the curves bandwidth and can be directly related to turbine solidity. Higher solidity often results in a narrower curve, being less efficient over a range of wind speeds, whilst lower solidity tends to produce a broader performance range.

APPENDIX D – MATLAB Code Validation

Basic mathematical validation of the codes output was conducted with the helical turbine (smallest swept area). The code estimates an energy production of around 150 kWh per year, with average wind speed of 4m/s, mechanical / electrical losses of around 5% and swept area of 2.53 m^2. The C_p-TSR curve has a max C_p of around 25%, an average C_p of half of the max (12.5%) is assumed to account for C_p not operating at its max over all wind speeds (effects of the Cp-TSR curve). The Weibull distribution is used as when wind speed increases wind power increases cubically, the occasional high wind speed events the turbine can take advantage of cause an increase in power. To account for this the effective mean velocity should be calculated using the n_{th} raw moment equation for the Weibull distribution (*Equation 0.1*) [92].

$$\overline{v^3} = C^3 * \Gamma\left(1 + \frac{3}{k}\right)$$

Equation 0.1

This equation uses $\overline{v^3}$ rather than $(\bar{v})^3$ therefore, accounting for the cubic weighting of higher wind speeds which contribute disproportionately to energy. The k parameter is 1.9 and C can be estimated as 4.5 m/s (explanation for these values in section 3.1.3). Using *Equation 0.1*, $\overline{v^3}$ can be calculated as 135m^3/s^3. Now the available wind energy over a year can be calculated:

$$P = \frac{1}{2}\,\rho\,\overline{v^3}\,A = \ 387\ W * Hours\ In\ a\ Year\ (8760) = \ 1{,}900\ kWh/year$$

Using the Weibull cumulative frequency distribution, the probability of wind speeds below the turbines cut-in (3 m/s) is [88]:

$$P(v < 3) = 1 - \exp\left[-\left(\frac{3}{c}\right)^k\right] \approx 37\%$$

Equation 0.2

To calculate the energy available above cut-in speed:

$$E_{above\ cut-in} = E_{total} * (1 - 0.37) = \ 1{,}200\ kWh/year$$

Multiply by average power coefficient and 5% mechanical / electrical losses:

$$E = \ 1{,}200 * C_{Pavr} * losses = \ 140\ kWh/year$$

This value agrees closely with the codes output of 150 kWh/year. The slight difference is likely due to the code more accurately representing the Cp-TSR curve, in this case as the helical turbine can operate over a slightly larger TSR range it likely has a higher average Cp.